

LECTURE ON EXPLICIT TRACE FORMULAE ON GL_3

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CONTENTS

1. Automorphic forms on general linear groups	2
1.1. General linear and special linear groups	2
1.2. The infinitesimal characters	3
1.3. Lattices	4
1.4. Examples of lattices	5
1.5. Automorphic forms on G_n	5
1.6. Cusp forms on G_n	7
2. Statement of the explicit trace formula of GL_3	8
2.1. Co-compact cases	8
3. The Selberg trace formula (The co-compact case)	10
3.1. Square integrable forms	10
3.2. Selberg trace formula (abstract form)	11
4. Harmonic analysis on $GL_n(\mathbb{R})$	11
4.1. Invariant distributions and their Fourier transforms	12
4.2. The case $G_1 = \mathbb{R}^\times$	12
4.3. Basic properties of the orbital integrals	13
4.4. Regular semisimple orbital integrals	13
4.5. The case GL_2	15
4.6. Tempered representations	17
4.7. Orbital integrals for general elements	18
5. Proof of the explicit trace formula	19
5.1. Harmonic analysis on GL_3	19
5.2. A sketch of the proof of Theorem 2.3	23
6. Trace formulae of non-compact quotients for GL_2	25
6.1. The spectral analysis of GL_2	25
6.2. The modified kernel	28
6.3. The spectral side	29
6.4. The geometric side	30
6.5. Arthur-Selberg trace formula	31
6.6. The weighted orbital integrals	31
6.7. The fine geometric expansion	33
6.8. Invariant trace formula with the fine geometric expansion	33
6.9. An invariant avatar of the weighted orbital integral	36
7. Towards an explicit trace formula of $SL_3(\mathbb{Z})$: work in progress	37
7.1. Further readings	37

Course description: The Arthur-Selberg trace formula is an important tool in the modern theory of automorphic forms. Initially, it was invented by A.Selberg in a general framework of weakly symmetric spaces; By applying an explicit version of the trace formula for the modular forms on the Poincare upper half plane, Selberg showed that there are infinitely many linearly independent Maass wave cusp forms on $SL_2(\mathbb{Z})$. Arthur developed the Selberg's idea in a framework of adelizations of reductive groups over number fields to serve the trace formula as an apparatus to construct the functorial transfers (conjectured by Langlands) of automorphic representations of different groups through comparison of the trace formulae. In this narrative, explicit forms of terms in the trace formulae are not cared about so far simply because they are not necessary. However, if one would like to apply the trace formula for computational problems in automorphic forms, one need to calculate terms in Arthur-Selberg's trace formula further. In this series of lectures, we focus on the group GL_3 , especially for uniform lattices of $SL_3(\mathbb{R})$, to explain the idea how an explicit version of the trace formula is deduced from an abstract trace formula. On the way, we give an introduction to the harmonic analysis on a real reductive groups explaining basic concepts like the global characters, the orbital integrals, etc. In the final part of the lecture, I would like to mention an on going project with Werner Hoffmann (Bielefeld) and Stoshi Wakatsuki (Kanazawa) on an explicit trace formula on $SL_3(\mathbb{Z})$. The lectures will cover the following topics:

- Automorphic forms on $GL_n(\mathbb{R})$.
- Basic harmonic analysis on $GL_n(\mathbb{R})$.
- Automorphic forms on $GL_2(\mathbb{A})$; Arthur-Selberg invariant trace formula.
- The weighted characters and the weighted orbital integrals.
- Towards an explicit trace formula for $SL_3(\mathbb{Z})$

Monday :

(I) 14:00 – 15:30 (1.5h) §1, §2, §3

(II) 15:50 – 17:20 (1.5h) §3, §4, §5

Tuesday :

(III) 10:40 – 11:30 (50min) §6,

(IV) 14:00 – 15: 30 (1.5h) §6, §7

Wednesday :

(V) 10:40 – 11: 30 (50min)

1. AUTOMORPHIC FORMS ON GENERAL LINEAR GROUPS

1.1. **General linear and special linear groups.** For $n \in \mathbb{Z}_{>0}$,

$$G_n := GL_n(\mathbb{R}),$$

$$G_n^+ := \{g \in G_n \mid \det g > 0\},$$

$$G_n^1 := \{g \in G_n \mid |\det g| = 1\}.$$

Then, G_n^+ is the connected component of the identity matrix 1_n , and $G_n^+\delta_n$ is the other connected component, where

$$\delta_n := \begin{cases} \text{diag}(1_{n-1}, -1) & (n \equiv 0 \pmod{2}), \\ -1_n & (n \equiv 1 \pmod{2}) \end{cases}$$

We have $G_n^1 = \text{SL}_n(\mathbb{R}) + \bigsqcup \text{SL}_n(\mathbb{R})\delta_n$, and the direct product decomposition:

$$G_n = \{z1_n \mid z > 0\} G_n^1.$$

The group G_n , endowed with the induced topology from the ambient Euclidean space $\text{Mat}_n(\mathbb{R})$, is a Lie group. We fix a Haar measure on G_n (For example

$$dg = |\det g|^{-n} \prod_{ij} dg_{ij}, \quad g = (g_{ij})_{ij} \in G_n$$

works.) Let A_n be the subgroup of all diagonal matrices in G_n and U_n the subgroup of all the upper-triangular unipotent matrices in G_n . Then we choose dg such that

$$\int_{G_n} f(g) dg = \int_{A_n} \int_{U_n} \int_{O(n)} f(uak) \delta(a)^{-1} da du dk,$$

where

$$da := \prod_{j=1}^n \frac{da_j}{|a_j|}, \quad \delta(a) := \prod_{j=1}^n |a_j|^{n-2j+1}, \quad a = \text{diag}(a_1, \dots, a_n) \in A_n,$$

$$du := \prod_{1 \leq i < j \leq n} du_{ij} \quad u = (u_{ij})_{ij} \in U_n$$

and dk denotes the Haar measure of the orthogonal group

$$O(n) := \{k \in \text{GL}_n(\mathbb{R}) \mid {}^t k k = 1_n\}$$

of total volume 1. The Haar measure on G_n is restricted to the open subgroup G_n^+ of G_n . We fix a Haar measure dh on G_n^1 by demanding

$$\int_{G_n} f(g) dg = \int_{G_n^1} \left(\int_{\mathbb{R}_{>0}} f(zh) \frac{dz}{z} \right) dh$$

for any integrable function f on G_n . The subscript n is abbreviated in most of the time, i.e., G , G^1 , A , N for G_n , G_n^1 , A_n , U_n .

1.2. The infinitesimal characters. Let $G = G_n$ and

$$\mathfrak{g} := \text{Lie}(G).$$

$\mathfrak{g}$ is identified with the space of $n \times n$ -matrices $\text{Mat}_n(\mathbb{R})$ with the Lie bracket $[X, Y] := XY - YX$ for two such matrices X, Y , by letting $X \in \text{Mat}_n(\mathbb{R})$ acts on the space of C^∞ -functions ϕ on G_n as

$$R(X)\phi(g) := \frac{d}{dt} \phi(g \exp(tX))|_{t=0}, \quad g \in G.$$

We usually extend the action of $\mathfrak{g}$ on functions linearly to the complexification $\mathfrak{g}^\mathbb{C} := \mathfrak{g} \otimes_\mathbb{R} \mathbb{C} = \text{Mat}_n(\mathbb{C})$. Then, we have the relation

$$R(X) \circ R(Y) - R(Y) \circ R(X) = R([X, Y]), \quad X, Y \in \mathfrak{g}^\mathbb{C}$$

of linear operators on $C^\infty(G)$. Let $U(\mathfrak{g}^\mathbb{C})$ be the universal enveloping algebra of $\mathfrak{g}^\mathbb{C}$. By the universality, there exists a unique algebra homomorphism $R : U(\mathfrak{g}^\mathbb{C}) \rightarrow \text{End}(C^\infty(G))$ extending the linear map $R : \mathfrak{g}^\mathbb{C} \rightarrow \text{End}(C^\infty(G))$.

Let $Z(\mathfrak{g}^{\mathbb{C}})$ be the center of $U(\mathfrak{g}^{\mathbb{C}})$. It is described by the Harish-Chandra homomorphism ([12, Chap VIII §5])

$$(1.1) \quad \gamma : Z(\mathfrak{g}^{\mathbb{C}}) \longrightarrow \mathbb{C}[H_1, \dots, H_n]^{S_n},$$

which is a $\mathbb{C}$ -algebra isomorphism from $Z(\mathfrak{g}^{\mathbb{C}})$ onto the algebra of all symmetric polynomials in H_j ($1 \leq j \leq n$), where S_n , the symmetric group of degree n , acts on the polynomial algebra $\mathbb{C}[H_1, \dots, H_n]$ by permutation of variables: $\sigma(H_j) = H_{\sigma^{-1}(j)}$. The map γ is constructed as follows. Set $\mathfrak{a} := \text{Lie}(A_n) = \sum_{j=1}^n \mathbb{C}E_{jj}$ and $\mathfrak{u} := \text{Lie}(U_n) = \sum_{1 \leq i < j \leq n} \mathbb{C}E_{ij}$ with $E_{ij} = (\delta_{it}\delta_{js})_{ts}$ the matrix units. Then, by using the Poincare-Birkhoff-Witt theorem, we can show that $U(\mathfrak{a}^{\mathbb{C}}) \cap U(\mathfrak{g}^{\mathbb{C}})\mathfrak{u} = \{0\}$ and $Z(\mathfrak{g}) \subset U(\mathfrak{a}^{\mathbb{C}}) + U(\mathfrak{g}^{\mathbb{C}})\mathfrak{u}$. Then, γ is the composite of the following maps:

$$Z(\mathfrak{g}^{\mathbb{C}}) \subset U(\mathfrak{a}^{\mathbb{C}}) \oplus U(\mathfrak{g}^{\mathbb{C}})\mathfrak{u} \xrightarrow{\text{Pr}_1} U(\mathfrak{a}^{\mathbb{C}}) \xrightarrow{\sigma} U(\mathfrak{a}^{\mathbb{C}}) \cong \mathbb{C}[H_1, \dots, H_n]$$

where $\sigma : U(\mathfrak{a}^{\mathbb{C}}) \rightarrow U(\mathfrak{a}^{\mathbb{C}})$ is the algebra isomorphism such that $\sigma(H) = H - \rho(H)1$ for $H \in \mathfrak{a}^{\mathbb{C}}$ with

$$\rho := \frac{1}{2} \sum_{j=1}^n (n - 2j + 1)e_j \in (\mathfrak{a}^{\mathbb{C}})^*$$

and $\{e_j\}_j \subset (\mathfrak{a}^{\mathbb{C}})^*$ the dual basis of $\{E_{ii}\}_i$.

Definition 1. A $\mathbb{C}$ -algebra homomorphism $\Xi : Z(\mathfrak{g}^{\mathbb{C}}) \rightarrow \mathbb{C}$ is called an *infinitesimal character*.

We have a specific element $C_G \in Z(\mathfrak{g}^{\mathbb{C}})$, called the Casimir element of G , defined by

$$C_G := \sum_j X_j X^j$$

where $\{X_j\}_j$ and $\{X^j\}_j$ any $\mathbb{C}$ -basis of $\mathfrak{sl}_n(\mathbb{R})$ such that $2^{-1}\text{tr}(X_j X^j) = \delta_{ij}$. Then, $\gamma(C_G) =$

Example of $\text{GL}_2(\mathbb{R})$: The matrices

$$Z = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, W = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \bar{W} = \begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}$$

form a $\mathbb{C}$ -basis of the Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ such that

$$[Z, W] = 2W, \quad [Z, \bar{W}] = -2\bar{W}, \quad [W, \bar{W}] = Z$$

In terms of this basis, we have

$$C_{G_2} = \frac{1}{2}Z^2 + W\bar{W} + \bar{W}W.$$

Let $C_1 := E_{11} + E_{22}$ the generator of the Lie algebra of the center of G_2 . Then $Z(\mathfrak{g}_2^{\mathbb{C}}) = \mathbb{C}[C_2, C_1]$.

1.3. Lattices. A subgroup Γ of G_n^1 is called a lattice if

- (a) Γ is discrete,
- (b) Γ admits a measurable fundamental set $F_\Gamma \subset G_n$ such that $\text{vol}(F_\Gamma) < \infty$.

We say that a lattice $\Gamma \subset G_n^1$ is uniform if the quotient space $\Gamma \backslash G_n^1$ is compact.

1.4. Examples of lattices. We only consider lattices from one of the following classes.

- (i) (Congruence subgroups): $\mathrm{SL}_n(\mathbb{Z})$ and $\mathrm{GL}_n(\mathbb{Z})$ are lattices of G_n^1 . More generally, if Γ contains one of subgroups $\Gamma^{(n)}(N) = \{\gamma \in \mathrm{SL}_n(\mathbb{Z}) \mid \gamma \equiv 1_n \pmod{N}\}$ ($N \in \mathbb{Z}_{>0}$), then Γ is a lattice of G_n^1 .
- (ii) Let A be a central division over $\mathbb{Q}$, i.e., A is a $\mathbb{Q}$ -algebra with the multiplicative unit 1_A whose center is $\mathbb{Q}1_A$, and A is a skew field. By a general theory, we have the reduced norm map $\mathrm{Nr}_A : A \rightarrow \mathbb{Q}$. Assume that A is split over $\mathbb{R}$, i.e., there exists $\mathbb{R}$ -algebra isomorphism

$$\iota_{\mathbb{R}} : A \otimes_{\mathbb{Q}} \mathbb{R} \longrightarrow \mathrm{Mat}_n(\mathbb{R}).$$

Suppose n is prime for simplicity from now on. Then, a central simple algebra over $\mathbb{Q}$ is either isomorphic to the matrix algebra $\mathrm{Mat}_n(\mathbb{Q})$ or a division algebra (skew-field). By class field theory, such A is constructed up to isomorphism as follows([18]): For each prime number p , up to isomorphisms, there exist exactly n central simple algebras $\mathfrak{A}/\mathbb{Q}_p$ of degree n parametrized by the finite additive group

$$\frac{1}{n}\mathbb{Z}/\mathbb{Z} = \left\{\frac{0}{n}, \frac{1}{n}, \dots, \frac{n-1}{n}\right\} \quad (\subset \mathbb{Q}/\mathbb{Z})$$

in such a way that $\frac{0}{n}$ corresponds to the matrix algebra $\mathrm{Mat}_n(\mathbb{Q}_p)$ and the remaining non-zero $\frac{i}{n}$ corresponds to non-isomorphic central division algebras over $\mathbb{Q}_p$. The invariant of $\mathfrak{A}/\mathbb{Q}_p$ is denoted by $\mathrm{inv}_p(\mathfrak{A}) (\in \frac{1}{n}\mathbb{Z}/\mathbb{Z})$. Then, given any collection $\{\mathfrak{A}_p\}_{p<\infty}$ of central simple algebras over p -adic fields such that $\mathrm{inv}_p(\mathfrak{A}_p) = 0$ for almost all p and $\sum_p \mathrm{inv}_p(\mathfrak{A}_p) = 0$, there exists up to isomorphisms unique central simple algebra A over $\mathbb{Q}$ which is split over $\mathbb{R}$ and $A \otimes_{\mathbb{Q}} \mathbb{Q}_p \cong \mathfrak{A}_p$. A being a central division algebra over $\mathbb{Q}$ means $\mathrm{inv}_p(A \otimes_{\mathbb{Q}} \mathbb{Q}_p) \neq 0$ for at least 2 primes p . Let $\mathcal{O} \subset A$ be an order of A , i.e., $\mathcal{O}$ is a subring of A free of rank n^2 over $\mathbb{Z}$. Then, by [6, Theorem 11.6], the image of

$$\mathcal{O}^1 := \{a \in \mathcal{O} \mid \mathrm{Nr}_A(a) = 1\}$$

under the map

$$\mathcal{O}^1 \subset A^\times \hookrightarrow (A \otimes_{\mathbb{Q}} \mathbb{R})^\times \xrightarrow{\iota_{\mathbb{R}}} \mathrm{Mat}_n(\mathbb{R})^\times = G_n$$

yields a uniform lattice $\Gamma_{\mathcal{O}}$ of G_n^1 .

1.5. Automorphic forms on G_n .

1.5.1. The classical cases. To motivate a general definition of automorphic forms, we start from the classical modular forms on the upper half-plane

$$\mathfrak{h} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}.$$

The group $G_2^+ = \mathrm{GL}_2(\mathbb{R})^+$ acts on $\mathfrak{h}$ by the Mobius transformation

$$g(\tau) := \frac{a\tau + b}{c\tau + d}, \quad g = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{R}), \tau \in \mathfrak{h}.$$

The action is transitive and the stabilizer of the imaginary unit $i \in \mathfrak{h}$ coincides with $\mathbb{R}_{>0} \mathrm{SO}(2)$, where

$$\mathrm{SO}(2) := \{k_\theta := \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \mid \theta \in [0, 2\pi)\},$$

so that

$$G_2^+/\mathbb{R}_{>0}\mathrm{SO}(2) \cong \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2) \cong \mathfrak{h}.$$

A holomorphic function $f(\tau)$ on $\mathfrak{h}$ is a modular form of weight l if

$$f(\gamma(\tau)) = (c\tau + d)^l f(\tau), \quad \gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \tau \in \mathfrak{h}$$

and it admits the q -expansion

$$f(\tau) = \sum_{n=0}^{\infty} a_n q^n, \quad q = e^{2\pi i \tau}.$$

(It is easy to see that non-zero modular forms $f(\tau)$ does not exists if $l < 0$ or $l > 0$ is odd.) Let

$$M_l(\mathrm{SL}_2(\mathbb{Z}))$$

denote the space of modular forms on $\mathfrak{h}$ of weight l .

For $f(\tau) \in M_l(\mathrm{SL}_2(\mathbb{Z}))$, we define a function ϕ_f on G_2 by setting

$$\phi_f(g) := (\det g)^{l/2} (c\tau + d)^{-l} f(g(i)), \quad g = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in G_2^+.$$

and $\phi_f(g) := \phi_f(\delta_2 g)$ if $g \in G_2 - G_2^+$. Then, $f \mapsto \phi_f$ yields a $\mathbb{C}$ -linear isomorphism from $M_l(\mathrm{SL}_2(\mathbb{Z}))$ onto the space of functions $\phi : G_2 \rightarrow \mathbb{C}$ satisfying the following conditions:

- (a) $\phi(\gamma g) = \phi_f(g)$ for $\gamma \in \mathrm{GL}_2(\mathbb{Z})$, $g \in G_2$.
- (b) $\phi(gz k_\theta) = e^{il\theta} \phi_f(g)$ for $k_\theta \in \mathrm{SO}(2)$, $z \in \mathbb{R}_{>0}$ and $g \in G_2$.
- (c) $R(\bar{W})\phi = 0$, where

$$\bar{W} := \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix} \in \mathfrak{sl}_2(\mathbb{C}).$$

- (d) There exists $m \in \mathbb{R}$ and $C > 0$ such that

$$|\phi(g)| \leq C \mathrm{tr}(g^t g)^m, \quad g \in G_2^1.$$

We have

$$R(C_{G_2})\phi_f = \frac{1}{2}l(l-2)\phi_f.$$

Since $\{z1_2 \mid z > 0\}$ acts trivially on ϕ by (b), the algebra $Z(\mathfrak{g}_2^{\mathbb{C}}) = \mathbb{C}[C_{G_2}, C_1]$ acts on ϕ by a character $\Xi : Z(\mathfrak{g}_2^{\mathbb{C}}) \rightarrow \mathbb{C}$:

$$R(D)\phi = \Xi(D)\phi, \quad D \in Z(\mathfrak{g}_2^{\mathbb{C}}).$$

1.5.2. General case. We work with $G = G_n$ with $n \geq 2$. Let (τ, V_τ) be an irreducible finite dimensional representation of $\mathrm{SO}(n)$ or $O(n)$. Let $\Xi : Z(\mathfrak{g}^{\mathbb{C}}) \rightarrow \mathbb{C}$ be a $\mathbb{C}$ -algebra homomorphism. Let $\Gamma \subset G^1$ be a lattice.

Definition 2. We say that a C^∞ -function $\phi : G \rightarrow \mathbb{C}$ is an automorphic form on Γ of weight τ and infinitesimal character Ξ if it satisfies the conditions:

- $\phi(\gamma g) = \phi(g)$ for all $\gamma \in \Gamma$ and $g \in G$,
- $\phi((z1_n)g) = \phi(g)$ for $z \in \mathbb{R}_{>0}$ and

$$R(e_\tau)\phi(g) := \int_{\mathrm{SO}(n)} \phi(gk) e_\tau(k^{-1}) dk = \phi(g)$$

for all $g \in G$, where $e_\tau(k) := (\dim \tau) \mathrm{tr} \tau(k^{-1})$ for k in $\mathrm{SO}(n)$ or $O(n)$.

- $R(D)\phi = \Xi(D)\phi$ for all $D \in Z(\mathfrak{g}^{\mathbb{C}})$.
- There exists $m \in \mathbb{R}$ and $C > 0$ such that $|\phi(g)| \leq C(\mathrm{tr}(g^t g))^m$ for all $g \in G^1$.

Let $\mathcal{A}(\Gamma; \tau, \Xi)$ denote the space of all such ϕ .

Lemma 1.1 (Harish-Chandra).

$$\dim_{\mathbb{C}} \mathcal{A}(\Gamma; \tau, \Xi) < \infty.$$

Let $\tau_l : \mathrm{SO}(2) \rightarrow \mathbb{C}^1$ be the character $k_\theta \mapsto e^{-il\theta}$.

Lemma 1.2. *The map $f \mapsto \phi_f$ yields a $\mathbb{C}$ -linear bijection from $M_l(\mathrm{SL}_2(\mathbb{Z}))$ onto the space $\mathcal{A}(\mathrm{GL}_2(\mathbb{Z}); \tau_l, \Xi_l)$.*

Remark 1. $\mathcal{A}(\mathrm{SL}_2(\mathbb{Z}); \tau_l, \Xi_l)$ is 2 times larger space than $\mathcal{A}(\mathrm{GL}_2(\mathbb{Z}); \tau_l, \Xi_l)$ in dimension.

Example (Maass forms) : The Maass form is another important class of modular forms on $\mathrm{SL}_2(\mathbb{Z})$, which are defined to be real analytic functions $f(x + iy)$ such that

- $f(\gamma(\tau)) = f(\tau)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$
- $-y^2(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})f(\tau) = \frac{s^2 - 1}{2}f(\tau)$ for some $s \in \mathbb{C}$.
- It satisfies an estimate $|f(x + iy)| \ll y^N$ on the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ for some N

Then, the map $f \mapsto \phi_f$ with $\phi_f(g) = \phi_f(\delta_2 g) := f(g(i))$ for $g \in G_2^+$ yields a linear bijection from the space of Maass forms of eigenvalue $\frac{s^2 - 1}{2}$ onto the space $\mathcal{A}(\mathrm{GL}_2(\mathbb{Z}); \tau_0, \Xi^{(s)})$, where $\Xi^{(s)} : Z(\mathfrak{g}_2^{\mathbb{C}}) \rightarrow \mathbb{C}$ is defined by $\Xi^{(s)}(C_{G_2}) = \frac{s^2 - 1}{4}$ and $\Xi^{(s)}(C_1) = 0$. A higher dimensional analogue of the Maass form is defined in the current framework as follows. For $g \in G_n^1$, let

$$g = y(g)x(g)k(g), \quad y(g) \in A_n, x(g) \in U_n, k \in \mathrm{SO}(n)$$

be the Iwasawa decomposition of g . The absolute value of entries of the element $y(g) = \mathrm{diag}(y_1(g), \dots, y_n(g))$ are uniquely determined. Then, for $s = (s_1, \dots, s_n) \in \mathbb{C}^n$, the function $y^{(s)}(g) := \prod_{j=1}^n |y_j(g)|^{s_j + n - 2j + 1}$ turns out to be an eigenfunction of $Z(\mathfrak{g}^{\mathbb{C}})$, i.e., there exists $\Xi^{(s)} : Z(\mathfrak{g}_n^{\mathbb{C}}) \rightarrow \mathbb{C}$ such that $R(D)y^{(s)} = \Xi^{(s)}(D)y^{(s)}$ for $D \in Z(\mathfrak{g}_n^{\mathbb{C}})$. Let τ_0 be the trivial representation of $\mathrm{SO}(n)$. Then, elements of the space $\mathcal{A}(\Gamma, \tau_0, \Xi^{(s)})$ are defined to be Maass forms on a lattice $\Gamma \subset G^1$.

1.6. Cusp forms on G_n .

- $G_n = \mathrm{GL}_n(\mathbb{R})$ ($n = 1, 2, \dots$).
- $A_n = \{\mathrm{diag}(t_1, t_2, \dots, t_n) \mid t_j \in \mathbb{R}^\times\}$.
- $U_n = \{u \in G_n \mid u_{ij} = 0 \ (1 \leq j < i \leq n)\}$.
- $B_n = T_n U_n$: the upper triangular Borel subgroup of G_n .
- $\underline{n} = (n_1, \dots, n_r)$: a partition of n , i.e., $n_j \geq 1$, $n = \sum_{j=1}^r n_j$.
- A closed subgroup containing the upper-triangular Borel B_n is called a standard parabolic subgroup of G_n . They are exhausted by

$$P_{\underline{n}} = \left\{ \begin{bmatrix} g_1 & * & * & \cdots & * \\ 0 & g_2 & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & g_r \end{bmatrix} \in G_n \mid g_j \in G_{n_j} \ (1 \leq j \leq r) \right\} = M_{\underline{n}} N_{\underline{n}}$$

with $\underline{n} = (n_1, \dots, n_r)$ a partition of n .

- $O(n)$: the orthogonal group of degree n , which is a maximal compact subgroup of G_n . Let Z_n be the center of G_n , i.e., Z_n is the set of scalar matrices in G_n .

A modular form $f(\tau) \in M_l(\mathrm{SL}_2(\mathbb{Z}))$ is a cusp form, i.e., $a(0) = 0$, if and only if

$$\int_{U_2 \cap \mathrm{SL}_2(\mathbb{Z}) \backslash U_2} \phi_f(ug) \, du = 0, \quad g \in G_2.$$

Definition 3. Let Γ be a congruence subgroup of G_n^1 . Then, an automorphic form $\phi \in \mathcal{A}(\Gamma; \tau, \Xi)$ is defined to be a cusp form if it satisfies the condition: For any non-trivial partition $\underline{n}$ of n ,

$$(1.2) \quad \int_{N_{\underline{n}} \cap \Gamma \backslash N_{\underline{n}}} \phi(ug) \, du = 0$$

for all g . Let $\mathcal{A}_0(\Gamma; \tau, \Xi)$ be the space of all cusp forms in $\mathcal{A}(\Gamma; \tau, \Xi)$.

For convenience, if $\Gamma = \Gamma_{\mathcal{O}}$ with an order $\mathcal{O}$ of a central division algebra A , we set $\mathcal{A}_0(\Gamma; \tau, \Xi) = \mathcal{A}(\Gamma; \tau, \Xi)$.

2. STATEMENT OF THE EXPLICIT TRACE FORMULA OF GL_3

2.1. Co-compact cases. Let A be a central division algebra of degree 3 over $\mathbb{Q}$ and let $\Gamma = \Gamma_{\mathcal{O}}$ be the uniform lattice of $\mathbf{SL}_3(\mathbb{R})$ associated with an order $\mathcal{O} \subset A$. Let $l \in \mathbb{Z}_{>0}$ and $\varepsilon \in \{0, 1\}$ the parity of $l + 1$. We are interested in the space of automorphic forms

$$\mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}) \quad (t \in \mathbb{R})$$

on $G^1 = \{\pm 1_3\} \mathrm{SL}_3(\mathbb{R})$, where $\tilde{\Gamma} = (\pm 1_3) \Gamma_A$, τ_l^+ an irreducible unitary representation of $\mathrm{O}(3)$ of dimension $2l + 1$ such that $\tau_{l+1}^+(-1_3) = \mathrm{Id}$ (which is unique up to isomorphisms), and $\Xi_{l,it} : Z(\mathfrak{g}^{\mathbb{C}}) \rightarrow \mathbb{C}$ is defined by

$$\Xi_{l,it}(D) = \gamma(D) \left(\frac{l}{2} + \frac{it}{3}, \frac{-l}{2} + \frac{it}{3}, \frac{-2it}{3} \right)$$

where $\gamma : Z(\mathfrak{g}^{\mathbb{C}}) \cong \mathbb{C}[H_1, H_2, H_3]^{S_3}$ the Harish-Chandra isomorphism. Define the set of spectral parameters

$$\Lambda_l := \{t \in \mathbb{R} \mid \mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}) \neq \{0\}\};$$

then, it is proved that Λ_l is a countable discrete subset of $\mathbb{R}$. Let

$$N_l^{\Gamma}(T) := \sum_{\substack{t \in \Lambda_l \\ |t| < T}} \dim_{\mathbb{C}} \mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}), \quad T > 0$$

be the counting function of spectral parameters up to $T > 0$. Then,

Proposition 2.1 (Weyl's law).

$$N_l^{\Gamma}(T) = \frac{3l}{\pi^3} \mathrm{vol}(\Gamma \backslash \mathrm{SL}_3(\mathbb{R})) T^3 + O_l(T^2) \quad (T \rightarrow \infty)$$

Let $\mathfrak{E}_3(\Gamma)$ be the set of Γ conjugacy classes in $\Gamma - \{1_3\}$; we fix a complete set of representatives in Γ of $\mathfrak{E}_3(\Gamma)$ once and for all. For $\eta \in \mathfrak{E}_3(\Gamma)$, we have that $\mathbb{Q}[\eta]$ is a cubic field extension of $\mathbb{Q}$ (Lemma 4.2). Let $\Phi_{\eta}(X)$ be the characteristic polynomial of η , i.e., $\Phi_{\eta}(X) := \det(X1_3 - \eta)$ and $\mathrm{Disc}(\Phi_{\eta})$ the discriminant of $\Phi_{\eta}(X)$; the polynomial $\Phi_{\eta}(X)$ which is the minimal polynomial of η over $\mathbb{Q}$ is an irreducible cubic polynomial of rational coefficients¹. There are two possibilities:

¹Actually, Φ_{η} is a polynomial with integer coefficients of constant term -1 , so that η is a unit of the integer ring of the complex cubic field $\mathbb{Q}(\eta)$

- (i) η is real, i.e., $\Phi_\eta(X)$ has three different real roots, so that $\mathbb{Q}(\eta)$ is a totally real cubic field.
- (ii) η is complex, i.e., $\Phi_\eta(X)$ has two non-real complex roots and one real roots, so that $\mathbb{Q}(\eta)$ is a complex cubic field, i.e., with one complex place and one real place).

Let $\mathfrak{E}_3(\Gamma)_{\mathbb{R}}$ and $\mathfrak{E}_3(\Gamma)_{\mathbb{C}}$ denotes the set of all real conjugacy classes and the set of all complex Γ -conjugacy classes, respectively. We focus on the complex case. Let r_γ be the unique real root of $\Phi_\gamma(X)$ for $\gamma \in \mathfrak{E}_3(\Gamma)_{\mathbb{C}}$; then $r_\gamma > 0$ and the complex roots of $\Phi_\gamma(X)$ are of the form $r_\gamma^{-1/2}e^{\sqrt{-1}\theta(\gamma)}$ and $r_\gamma^{-1/2}e^{-\sqrt{-1}\theta(\gamma)}$ with $\theta(\gamma) \in [0, 2\pi) - \{\pi\}$. We define

$$\mathcal{L}(\gamma) := \frac{1}{2} |\log r_\gamma|.$$

Note that

$$|\text{Disc}(\Phi_\gamma)|^{1/2} = 4 \sin |\theta(\gamma)| (\cosh 3\mathcal{L}(\gamma) - \cos \theta(\gamma)).$$

We easily see that $\mathcal{L}(\gamma)$ and $|\text{Disc}(\Phi_\gamma)|$ are invariant by $\gamma \rightarrow \gamma^{-1}$. This motivate us to consider the quotient set $\mathfrak{E}_3(\Gamma)_{\mathbb{C}}^\pm$ of $\mathfrak{E}_3(\Gamma)_{\mathbb{C}}$ by the equivalence relation identifying γ and γ^{-1} up to Γ -conjugacy.

Lemma 2.2. *For $\gamma \in \mathfrak{E}_3(\Gamma)_{\mathbb{C}}^\pm$, there exists a unique primitive element γ_0 such that $\gamma = \gamma_0^n$ ($n \geq 1$). The centralizer Γ_γ is an infinite cyclic group generated by γ_0 .*

Recall the Chebyshev polynomial of the second kind $U_n(x)$ ($n \in \mathbb{Z}_{\geq 0}$). It may be defined as a unique polynomial in x such that

$$\frac{\sin((n+1)\theta)}{\sin \theta} = U_n(\cos \theta), \quad \theta \in (-\pi, \pi) - \{0\}.$$

Theorem 2.3. *Let $l \in \mathbb{Z}_{>0}$ and $\varepsilon \in \{0, 1\}$ such that $l+1 \equiv \varepsilon \pmod{2}$. For any function $\hat{h}(r)$ defined in a neighborhood of the strip $|\text{Im}(r)| < 3/2$ satisfying*

- $\hat{h}(r)$ is holomorphic on $|\text{Im}(r)| < 3/2 + \delta$ for some $\delta > 0$.
- there exists a constant $C > 0$ such that

$$|\hat{h}(r)| \leq C(1 + |\text{Re}(r)|)^{-3-\delta}, \quad (|\text{Im}(r)| < 3/2 + \delta)$$

Set

$$(2.1) \quad h(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(r) e^{-iur} dr, \quad u \in \mathbb{R}.$$

Then we have the identity

$$\begin{aligned} & \sum_{t \in \Lambda_l} \dim_{\mathbb{C}} \mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}) \cdot \hat{h}(t) - \delta_{l,2} \hat{h}(0) \\ &= - \sum_{\gamma \in \mathfrak{E}_3(\Gamma)_{\mathbb{C}}^\pm} \frac{\mathcal{L}(\gamma_0)}{\cosh 3\mathcal{L}(\gamma) - \cos \theta(\gamma)} U_{l-1}(\cos \theta(\gamma)) h(2\mathcal{L}(\gamma)) \\ & \quad + \frac{1}{4\pi^3} \text{vol}(\Gamma \backslash \text{SL}_3(\mathbb{R})) \int_{\mathbb{R}} l \left(t^2 + \frac{l^2}{4} \right) \hat{h}(t) dt. \end{aligned}$$

with all series and integrals being absolutely convergent.

Remark 2. We have an explicit trace formula for Maass forms on $\tilde{\Gamma}$ similarly, which equates the spectral sum

$$\sum_{(s_1, s_2)} \dim_{\mathbb{C}} \mathcal{A}(\tilde{\Gamma}; \tau_0, \Xi^{(s_1, s_2, -s_1-s_2)}) \hat{h}(s_1, s_2)$$

depending on 2 variable test function $\hat{h}(s_1, s_2)$ with an infinite series over $\gamma \in \mathfrak{E}_3(\Gamma)_{\mathbb{R}}$ (totally real cubic fields) as well as a series over $\gamma \in \mathfrak{E}_3(\Gamma)_{\mathbb{C}}$ (complex cubic fields).

Remark 3. Any central simple division algebra A of degree 3 over $\mathbb{Q}$ automatically splits at the archimedean place, i.e., $A \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbf{Mat}_3(\mathbb{R})$. D. Flath in his thesis ([13]) established a transfer from automorphic representations on $A^{\times}$ for such A to cuspidal automorphic representations on GL_3 by a comparison of trace formulas, following the method originally due to Jacquet-Langlands for GL_2 . In this work, no explicit formula of terms are necessary. Main issue is transfer of orbital integrals over p -adic fields.

If we introduce Hecke operators on our space $\mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l, it})$ to decompose it into Hecke eigenspaces, each eigenform corresponds to an irreducible cuspidal automorphic representation $\Pi = \otimes_p \Pi_p$ of $\mathrm{GL}_3(\mathbb{A})$ whose local component Π_p at a ramified prime p (i.e., $A \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is a division algebra) is an essentially square-integrable representation of $\mathrm{GL}_3(\mathbb{Q}_p)$.

3. THE SELBERG TRACE FORMULA (THE CO-COMPACT CASE)

Let $G = G_n$ ($n \geq 2$).

3.1. Square integrable forms. Let $\Gamma \subset G^1$ is a lattice.

Definition 4. $\phi \in \mathcal{A}(\Gamma; \tau, \Xi)$ is a square-integrable form if

$$\int_{\Gamma \backslash G^1} |\phi(g)|^2 dg < \infty.$$

The space of all such ϕ is denoted by $\mathcal{A}_{(2)}(\Gamma; \tau, \Xi)$.

We have the inclusion

$$\mathcal{A}_0(\Gamma; \tau, \Xi) \subset \mathcal{A}_{(2)}(\Gamma; \tau, \Xi).$$

When, Γ is of congruence type, let $L_0^2(\Gamma \backslash G^1)$ be the space of all square integrable functions on $\Gamma \backslash G^1$ that satisfies the condition (1.2) for all non-trivial partitions $\underline{n}$ of n and for almost all $g \in G^1$. When Γ is from a division algebra, we set $L_0^2(\Gamma \backslash G^1) = L^2(\Gamma \backslash G^1)$. Then, by $G^1 \cong \mathbb{R}_{>0} \backslash G$, the group G acts on $L_0^2(\Gamma \backslash G^1)$ by the right-regular representation, say R_{Γ} . Let

$$\Pi_u(G) \quad (\text{the unitary dual of } G)$$

be the set of equivalence classes of irreducible unitary representations of G .

Theorem 3.1 (Gelfand-Graev-Piatetski-Schapiro). *The unitary representation R_{Γ} discretely decomposes to a countable sum of irreducible representations of G with finite multiplicities:*

$$L_0^2(\Gamma \backslash G^1) \cong \widehat{\bigoplus_{\pi \in \Pi_u(G)} m_{\Gamma}(\pi) \pi}, \quad m_{\Gamma}(\pi) \in \mathbb{Z}_{\geq 0}$$

For an irreducible unitary representation (π, H_{π}) , a vector $v \in H_{\pi}$ is said to be a smooth vector of π if the function $G \ni g \mapsto \pi(g)v \in H_{\pi}$ is C^{∞} ; let H_{π}^{∞} denote the set of all smooth vectors of π . There exists a $\mathbb{C}$ -algebra homomorphism $\Xi_{\pi} : Z(\mathfrak{g}^{\mathbb{C}}) \rightarrow \mathbb{C}$, called the infinitesimal character of π , such that

$$\pi(D)v = \Xi_{\pi}(D)v, \quad v \in H_{\pi}^{\infty}.$$

Then,

$$(3.1) \quad \dim_{\mathbb{C}} \mathcal{A}_0(\Gamma; \tau, \Xi) = \sum_{\substack{\pi \in \Pi_u(G) \\ \Xi_\pi = \Xi}} [\pi : \tau] m_\Gamma(\pi),$$

where $[\pi : \tau]$ is the multiplicity of τ in $\pi|_{\mathrm{SO}(n)}$. It is a theorem by Harish-Chandra that any $\pi \in \Pi_u(G)$ is admissible, i.e., any irreducible representation τ of $\mathrm{SO}(n)$ occurs in the restriction $\pi|_{\mathrm{SO}(n)}$ with finite multiplicity, i.e., $[\pi : \tau] < \infty$ ([12]).

3.2. Selberg trace formula (abstract form).

Definition 5. Let (π, H_π) be a unitary representation of G . For $f \in C_c^\infty(G)$, we have a bounded linear operator $\pi(f) : H_\pi \rightarrow H_\pi$ defined by the Bochner integral:

$$\pi(f)v = \int_G f(g)\pi(g)v \, dg, \quad v \in H_\pi.$$

When $\pi(f)$ is of trace class, we set

$$\widehat{f}(\pi) := \mathrm{trace}(\pi(f)).$$

It is a theorem by Harish-Chandra ([12, Theorem 10.2]) that $\pi(f)$ for $f \in C_c^\infty(G)$ is of trace class if π is irreducible and we have a continuous linear functional

$$\Theta_\pi : C_c^\infty(G) \ni f \mapsto \widehat{f}(\pi) \in \mathbb{C}$$

called the global character of π .

In this section, we consider a lattice $\Gamma = \Gamma_{\mathcal{O}}$ of G^1 arising from an order $\mathcal{O}$ in a central division algebra over $\mathbb{Q}$, so that $\Gamma \backslash G^1$ is compact. Then,

Proposition 3.2. For any $f \in C_c^\infty(G)$, the operator $R_\Gamma(f)$ on $L^2(\Gamma \backslash G^1)$ is of trace class, and

$$\mathrm{tr}(R_\Gamma(f)) = \int_{\Gamma \backslash G^1} K_f(g, g) \, dg,$$

where $K_f(g, g)$ is the diagonal restriction of a continuous function K_f on $\Gamma \backslash G^1 \times \Gamma \backslash G^1$ defined by

$$K_f(x, y) := \sum_{\gamma \in \Gamma} \int_{\mathbb{R}^\times} f(zx^{-1}\gamma y) \, dz, \quad x, y \in G.$$

Selberg trace formula :

For $f \in C_c^\infty(G)$, the trace $\mathrm{tr}(R_\Gamma(f))$ equals

$$(3.2) \quad \sum_{\pi \in \Pi_u(G)} m_\Gamma(\pi) \widehat{f}(\pi) = \sum_{\{\gamma\}_\Gamma} \mathrm{vol}(\Gamma_\gamma \backslash G_\gamma) \mathrm{Orb}(\gamma; f),$$

where, for each $\gamma \in \Gamma$, we fix a Haar measure on $G_\gamma := \{g \in G \mid g\gamma = \gamma g\}$ and use the quotient measure on $G_\gamma \backslash G$ to define the orbital integral

$$(3.3) \quad \mathrm{Orb}(\gamma; f) := \int_{G_\gamma \backslash G} f(g^{-1}\gamma g) \, dg$$

and the volume $\mathrm{vol}(\Gamma_\gamma \backslash G_\gamma)$.

4. HARMONIC ANALYSIS ON $\mathrm{GL}_n(\mathbb{R})$

We continue to work with $G = G_n$ ($n \geq 1$).

4.1. Invariant distributions and their Fourier transforms. The global character Θ_π of an irreducible unitary representation π is conjugation invariant, i.e., it satisfies

$$\Theta_\pi(f^g) = \Theta_\pi(f), \quad f \in C_c^\infty(G), \quad g \in G$$

where $f^g(x) := f(g^{-1}xg)$. In general, a continuous linear functional T on $C_c^\infty(G)$ is called to be an invariant distribution if it is conjugation-invariant, i.e., $T(f^g) = T(f)$ for all $f \in C_c^\infty(G)$ and $g \in G$. Another important class of invariant distribution is the orbital integral (3.3) associated to a G -conjugacy class $\{\gamma\}_G$ of $\gamma \in G$. It is a theorem by Rao [14] that the integral converges absolutely and defines a Radon measure on G . (When γ is semi-simple, then the orbit $\{\gamma\}_G$ is a closed subset of G , so that the convergence of the integral is obvious.)

Toy model :

Let G be a finite group and $L(G)$ the space of all $\mathbb{C}$ -valued functions on G . Let $\mathcal{I}(G) \subset L(G)$ be the subspace of all class functions ϕ , i.e., $\phi(g^{-1}xg) = \phi(x)$, ($\forall g, x \in G$). It is well known that $\mathcal{I}(G)$ has two natural basis.

- The set of *irreducible characters* $\chi_\rho(g) := \text{tr}(\rho(g))$,
- The set of characteristic functions $\mathbb{1}_C$ of *conjugacy classes* C in G .

For $f \in L(G)$, and an element $\gamma \in C$, the paring

$$\langle \mathbb{1}_C, f \rangle := \sum_{g \in G} \mathbb{1}_C(g) f(g) = \sum_{g \in C} f(g) = \sum_{g \in G_\gamma \backslash G} f(g^{-1}\gamma g)$$

is an obvious analogue of the orbital integral. The function $\mathbb{1}_C$ is a linear combination of irreducible characters χ_ρ .

Naive expectation :

The set of global characters $\{\Theta_\pi\}_\pi$ and the set of orbital integrals $\{\text{Orb}(\{\gamma\}_G; \cdot)\}$ both might contain spanning sets of the space of invariant distributions on G , so that we could express the orbital integral as a superposition of global characters (and vice versa):

$$(4.1) \quad \text{Orb}(\gamma; f) = \int_{\Pi_u(G)} \widehat{f}(\pi) a(\gamma, \pi) d\pi.$$

The expanding coefficients $\{a(\gamma, \pi)\}_\pi$ is referred to as the Fourier transform of $\text{Orb}(\gamma; \cdot)$ at π .

4.2. The case $G_1 = \mathbb{R}^\times$. In the abelian group $G = \mathbb{R}^\times$, two points are conjugate if and only if they are the same, i.e., a conjugacy class of $\gamma \in G$ is $\{\gamma\}$ and $G_\gamma = G$, so that $\text{Orb}(\gamma; f) = f(\gamma)$ for $f \in C_c^\infty(G)$. Formula (4.1) is nothing but the Fourier inversion formula of $\mathbb{R}^\times$:

$$f(\gamma) = \frac{1}{2} \sum_{\varepsilon \in \{0,1\}} \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(\chi_{\varepsilon, it}) \chi_{\varepsilon, it}(\gamma)^{-1} dt$$

where

$$\widehat{f}(\chi) := \int_{\mathbb{R}^\times} f(x) \chi(x) dx, \quad \chi \in \Pi_u(\mathbb{R}^\times).$$

4.3. Basic properties of the orbital integrals. For a while, we fix $P = P_{\underline{n}} = MN$, a standard parabolic subgroup of $G = G_n$, to work with. Set $K := O(n)$.

Definition 6. The group M acts on $\mathfrak{n} := \text{Lie}(N)$ by $\text{Ad}(m)|_{\mathfrak{n}} : X \mapsto mXm^{-1}$. Define a group homomorphism $\delta_P : M \rightarrow \mathbb{R}^\times$ by

$$\delta_P(m) := |\det(\text{Ad}(m)|_{\mathfrak{n}})|, \quad m \in M.$$

This is called the modulus character of P . Let dn be a Haar measure on N ; then $d(mnm^{-1})/dn = \delta_P(m)$.

If $\underline{n} = (n_1, \dots, n_q)$, then $m = \text{diag}(m_1, \dots, m_q)$ with $m_j \in G_{n_j}$ and

$$\delta_P(m) = \prod_{j=1}^q |\det m_j|^{\delta(j)}$$

with

$$\delta(j) := n - 2 \sum_{\nu=1}^{j-1} n_\nu + n_j, \quad (1 \leq j \leq q).$$

Definition 7. For $f \in C_c^\infty(G)$, define

$$f_P(m) := \delta_P(m)^{1/2} \int_K \int_N f(k^{-1}mnk) dn dk, \quad m \in M$$

By the Iwasawa decomposition $G = MNK$, we easily see that $f_P \in C_c^\infty(M)$.

Definition 8. Let $P = MN$ be a standard Levi subgroup of G . For a $\gamma \in M$ with the Jordan-Chevalley decomposition $\gamma = \gamma_u \gamma_\sigma = \gamma_\sigma \gamma_u$ and $f \in C_c^\infty(M)$, we set

$$I^M(\gamma, f) := |D_M(\gamma)|^{1/2} \text{Orb}^M(\gamma; f)$$

with

$$D_M(\gamma) := \det(1 - \text{Ad}(\gamma)|_{\mathfrak{m}/\mathfrak{m}_{\gamma_\sigma}}),$$

and

$$\text{Orb}^M(\gamma; f) := \int_{M_\gamma \backslash M} f(m^{-1}\gamma m) dm,$$

By properties of the Jordan-Chevalley decomposition, $M_\gamma = M_{\gamma_\sigma} \cap M_{\gamma_u}$.

Lemma 4.1 (Descent formula). Let $\gamma \in M$ be such that $G_\gamma \subset M$. Let $f \in C_c^\infty(G)$. Then,

$$I^G(\gamma; f) = I^M(\gamma; f_P)$$

4.4. Regular semisimple orbital integrals.

Definition 9. An element $g \in G_n$ is referred to as being regular semisimple if it has n distinct eigenvalues, or equivalently, the characteristic polynomial of g has no multiple roots. The associated orbital integrals $\text{Orb}(g, \cdot)$ are called the regular semi-simple orbital integrals.

Lemma 4.2. Let n be a prime and A a central division algebra of degree n . Let $\Gamma_{\mathcal{O}}$ be the uniform lattice of G_n^1 associated with an order $\mathcal{O} \subset A$. Then the set $\Gamma_{\mathcal{O}} - Z_n$ consists of regular semisimple elements of G_n .

Proof. Let $a \in A$. Obviously, $\mathbb{Q}1_A \subset \mathbb{Q}[a] \subset A$. As $\mathbb{Q}[a]$ is a subring of a division algebra A , it is a commutative field. We view A as a vector space over $\mathbb{Q}[a]$. Since $\dim_{\mathbb{Q}}(A) = n^2$ is a square of a prime and $\dim_{\mathbb{Q}} \mathbb{Q}[a]$ has to be a divisor of $\dim_{\mathbb{Q}}(A)$, we are forced to have $\dim_{\mathbb{Q}} \mathbb{Q}[a] \in \{1, n, n^2\}$. If a is non-central, then $\mathbb{Q}[a] \neq A$ and $\mathbb{Q}1_A \neq \mathbb{Q}[a]$, which means $\dim_{\mathbb{Q}} \mathbb{Q}[a] = n$. As $\mathbb{Q}[a]$ is a field extension of degree n , the minimal polynomial $P(X)$ of a over $\mathbb{Q}$ is a $\mathbb{Q}$ -irreducible polynomial of degree n , so that it has n distinct roots. An $\mathbb{R}$ -algebra isomorphism $\iota_{\mathbb{R}} : A \otimes_{\mathbb{Q}} \mathbb{R} \rightarrow \text{Mat}_n(\mathbb{R})$ is implicit in the definition of $\Gamma_{\mathcal{O}}$. Since $P(X)$ is the characteristic polynomial of the matrix $\iota_{\mathbb{R}}(a)$, we are done. $\square$

By the theory of Jordan canonical form, any regular semisimple element γ of G is $\text{GL}_n(\mathbb{C})$ -conjugate to a diagonal matrix

$$(4.2) \quad \text{diag}(\mu_1, \bar{\mu}_1, \dots, \mu_p, \bar{\mu}_p, \lambda_1, \dots, \lambda_p) \quad (n = 2q + p)$$

with q non-real complex conjugate pairs eigenvalues $\mu_j, \bar{\mu}_j$ and p real eigenvalues λ_i . By the regularity of γ , there is no repetition in the diagonal entries of (4.2); as such, the centralizer of (4.2) coincides with the group of diagonal matrices in $\text{GL}_n(\mathbb{C})$.

Lemma 4.3. *If $g_1, g_2 \in G$ are $\text{GL}_n(\mathbb{C})$ -conjugate, then g_1 and g_2 are G -conjugate.*

Set $\mu_j = r_j e^{\sqrt{-1}\theta_j}$ with $r_j > 0$ and $\theta_j \in [0, 2\pi)$. Then, γ is G -conjugate to

$$\text{diag}(r_1 k_{\theta_1}, \dots, r_q k_{\theta_q}, \lambda_1, \dots, \lambda_p).$$

and the centralizer in G of this matrix is

$$\text{diag}(\mathbb{R}^{\times} SO(2), \dots, \mathbb{R}^{\times} SO(2), \mathbb{R}^{\times}, \dots, \mathbb{R}^{\times})$$

which is contained in the standard Levi subgroup $M_{2, \dots, 2, 1, \dots, 1} \subset G_n$. Hence, Lemma 4.1 implies that regular semisimple orbital integrals on G is written in terms of orbital integrals of $f_{P_{2, \dots, 2, 1, \dots, 1}}$ on $M_{2, \dots, 2, 1, \dots, 1} = (G_2)^q \times (\mathbb{R}^{\times})^p$. Due to this, the regular semisimple orbital integrals on $\text{GL}_2(\mathbb{R})$ is of basic importance. We shall describe a rigorous version of the expansion formula (4.1) for regular orbital integrals; for this, we need to speak a little bit more of representations.

Definition 10 (Parabolically induced representations). *Let $P = MN$ be a standard parabolic subgroup of G . Let (σ, V_{σ}) be a continuous Hilbert representation of M . We define $\mathcal{K}(\sigma)$ to be the space of all those V -valued measurable functions $f : G \rightarrow V$ that satisfies*

$$f(nmg) = \delta_P(m)^{1/2} \sigma(m)(f(g)), \quad n \in N, m \in M$$

for almost everywhere in $g \in G$, with the Hermitian inner product

$$(f|f') := \int_K (f(k)|f'(k))_{\sigma} dk.$$

where $(\cdot|\cdot)_{\sigma}$ is the given inner product of V . We let the group G act on $\mathcal{K}(\sigma)$ by the right translations. The continuous Hilbert representation so obtained is symbolically denoted by σ^G . If σ is unitary, then the inner product of $\mathcal{K}(\sigma)$ is G -invariant, i.e., σ^G is unitary.

Notation : Let $\underline{n} = (n_1, \dots, n_t)$ be a partition of n . For a t -tuple $(\sigma_1, \dots, \sigma_t)$ with σ_j a continuous Hilbert representation of G_{n_j} , let

$$\sigma = \sigma_1 \boxtimes \dots \boxtimes \sigma_t$$

denote the representation of $M_{\underline{n}}$ on $V_{\sigma_1} \otimes \dots \otimes V_{\sigma_t}$ defined by $\sigma(m) := \sigma_j(m_1) \otimes \dots \otimes \sigma_t(m_t)$ for $m = \text{diag}(m_1, \dots, m_t) \in M_{\underline{n}}$

The following is a unique property of the general linear groups GL_n .

FACT : If σ is an irreducible unitary representation of M , then σ^G is *irreducible* unitary representation of G .

Lemma 4.4 (Formula of induced characters). *Suppose σ has a global character. Then σ^G also has a global character given in terms of the global character of σ as*

$$\widehat{f}(\sigma^G) = \widehat{f_P}(\sigma).$$

4.5. The case GL_2 . Any quasi-character (=continuous group homomorphism to $\mathbb{C}^\times$) of $\mathbb{R}^\times$ is uniquely written in the form

$$\chi_{\varepsilon,s} : t \mapsto |t|^s \mathrm{sgn}(t)^\varepsilon$$

with $\varepsilon \in \{0, 1\}$ and $s \in \mathbb{C}$. We have

$$\Pi_u(\mathbb{R}^\times) = \{\chi_{\varepsilon,s} \mid \varepsilon \in \{0, 1\}, s \in i\mathbb{R}\}.$$

For a representation (π, V) of G_2^+ , we define

$$V[n] := \{v \in V \mid \pi(k_\theta)v = e^{in\theta}v\}, \quad n \in \mathbb{Z}$$

The algebraic sum $V_{SO(2)} := \sum_{n \in \mathbb{Z}} V[n]$ is direct and invariant by the action of $SO(2)$; when π is a representation of G_2 , the element $\mathrm{diag}(1, -1) \in G_2$ flips $V[n]$ to $V[-n]$. Any vector $V_{SO(2)}$ is referred to as a $SO(2)$ -finite vector. The space $V_{SO(2)}$ is a $(\mathfrak{g}_2^\mathbb{C}, SO(2))$ -module.

4.5.1. Principal series representations. For a pair of quasi-characters (χ_1, χ_2) , we have defined a continuous Hilbert representation $\chi_1 \times \chi_2 := (\chi_1 \boxtimes \chi_2)^{G_2}$ on a Hilbert space $\mathcal{K}(\chi_1, \chi_2)$ to be the space of all measurable functions $f : G_2 \rightarrow \mathbb{C}$ that satisfies

$$f\left(\begin{bmatrix} t_1 & * \\ 0 & t_2 \end{bmatrix} g\right) = \chi_1(t_1)\chi_2(t_2)|t_1/t_2|^{1/2}f(g), \quad \forall \begin{bmatrix} t_1 & * \\ 0 & t_2 \end{bmatrix} \in P_{1,1}$$

for almost all $g \in G_2$. The representation $\chi_1 \times \chi_2$ is unitary if $\chi_1, \chi_2 \in \Pi_u(\mathbb{R}^\times)$. The center Z_2 acts on $\chi_1 \times \chi_2$ by $z1_2 \mapsto \chi_1\chi_2(z)$. If χ_1, χ_2 are unitary, then we have that $\chi_1 \times \chi_2$ is an irreducible unitary representation, and that $\chi_1 \times \chi_2$ and $\chi_2 \times \chi_1$ are unitary equivalent. The $O(2)$ -finite vectors $(\chi_1 \times \chi_2)_{O(2)}$ decomposes to a sum of characters $\varepsilon_n : k_\theta \rightarrow e^{-in\theta}$ of $SO(2)$ as

$$(\chi_1 \times \chi_2)_{SO(2)} \cong \bigoplus_{\substack{n \in \mathbb{Z} \\ \chi_1\chi_2(-1)=(-1)^n}} \varepsilon_n.$$

4.5.2. Discrete series representations. Let $l \in \mathbb{Z}_{>0}$ and $it \in i\mathbb{R}$. Let $\mathcal{D}_l$ denote the space of all those holomorphic functions $f : \mathbb{C} - \mathbb{R} \rightarrow \mathbb{C}$ that satisfies

$$\int_{\mathbb{C}-\mathbb{R}} |f(\tau)|^2 (\mathrm{Im}(\tau))^{l+1} d\mu(\tau) < \infty,$$

where $\mu(\tau) := \frac{dx dy}{y^2}$ for $\tau = x + iy$. We let the group $G_2 = G_2^+ \sqcup G_2 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ act on $\mathcal{D}_l$ by².

$$[\sigma_{l,it}(zg_1)f](\tau) := \chi_{\varepsilon,it}(z)(c\tau + d)^{-(l+1)}f\left(\frac{a\tau+b}{c\tau+d}\right), \quad g_1 \in \mathrm{SL}_2(\mathbb{R}), z \in \mathbb{R}^\times,$$

where $\varepsilon \in \{0, 1\}$ the parity of $l + 1$, and

$$[\sigma_{l,it}(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix})f](\tau) := \overline{f(\bar{\tau})}$$

²By $G_2^+ = Z_2\mathrm{SL}_2(\mathbb{R})$ and $Z_2 \cap \mathrm{SL}_2(\mathbb{R}) = \{\pm 1\}$, this defines a representation of G_2^+ only under the assumption $\chi_{\varepsilon,it}(-1) = 1$

The center acts on $\sigma_{l,it}$ by the character $\chi_{\varepsilon,it}$. It is known that $\sigma_{l,it}$ is an irreducible unitary representation of G_2 . Let $\mathcal{D}_l^\pm$ denote the subspace of $f \in \mathcal{D}_l$ such that $f(\tau) = 0$ ($\pm \text{Im}(\tau) < 0$). Then, $\mathcal{D}_l^\pm$ are irreducible G_2^+ -invariant subspaces such that $\mathcal{D}_l = \mathcal{D}_l^+ \oplus \mathcal{D}_l^-$. We have

$$(\mathcal{D}_l^+)_{SO(2)} = \bigoplus_{n \in \mathbb{Z}, l+1 \leq n} \varepsilon_n, \quad (\mathcal{D}_l^-)_{SO(2)} = \bigoplus_{n \in \mathbb{Z}, n \leq -(l+1)} \varepsilon_n.$$

We have an exact sequence of $(\mathfrak{g}_2^\mathbb{C}, O(2))$ -modules

$$(4.3) \quad 0 \longrightarrow (\sigma_{l,0})_{SO(2)} \longrightarrow (\chi_{\varepsilon,l/2} \times \chi_{0,-l/2})_{SO(2)} \longrightarrow F_{l-1} \longrightarrow 0$$

where $\varepsilon \equiv l+1 \pmod{2}$, $F_l = \text{sym}^l(\mathbb{C}^2) | \det |^{l/2}$ ($l > 0$) is the symmetric tensor representation of the tautological representation of G_2 on $\mathbb{C}^2$ twisted by the character $| \det |^{l/2}$.

For a character $\chi \in \Pi_u(\mathbb{R}^\times)$ such that $\chi(-1) = 1$ and $\varepsilon \in \{0, 1\}$, let $\mathbf{1}^\varepsilon[\chi]$ denote the one dimensional representation of G_2 defined by $zg_1 \mapsto \chi(z) \text{sgn}^\varepsilon(\det g_1)$ with $z \in \mathbb{R}^\times$ and $g_1 \in G_2^1$. Any 1-dimensional unitary representation is written like this.

Theorem 4.5. *The set $\Pi_u(G_2)$ is exhausted by the following representations:*

- *The principal series representations:*

$$\chi_1 \times \chi_2, \quad (\chi_1, \chi_2 \in \Pi_u(\mathbb{R}^\times))$$

- *The essentially square integrable representations:*

$$\sigma_{l,it}, \quad (l \in \mathbb{Z}_{\geq 0}, it \in i\mathbb{R})$$

- *Complementary representations:*

$$\chi_{\varepsilon,t+i\sigma} \times \chi_{\varepsilon,-t+i\sigma}, \quad (\sigma \in \mathbb{R}, t \in (0, 1/2), \varepsilon \in \{0, 1\}).$$

- *The one dimensional representations:*

$$\mathbf{1}_{G_2}^\varepsilon[\chi], \quad (\chi \in \Pi_u(\mathbb{R}^\times), \chi(-1) = 1, \varepsilon \in \{0, 1\}).$$

The only equivalence between these representations are

$$\chi_1 \times \chi_2 \cong \chi_2 \times \chi_1 \quad (\chi_1, \chi_2 \in \Pi_u(G_2))$$

Remark 4. *The representation $\chi_{1,it/2} \times \chi_{0,it/2}$ is also denoted by $\sigma_{0,it}$ and called the limit of discrete series representation.*

Proposition 4.6. *Let $a = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$ with $a_1 \neq a_2$. Then, for $f \in C_c^\infty(G_2)$*

$$(4.4) \quad I^G(a; f) = \left(\frac{1}{2\pi}\right)^2 \iint_{\mathbb{R}^2} \frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 \in \{0,1\}} \widehat{f}(\chi_{\varepsilon_1,it_1} \times \chi_{\varepsilon_2,it_2}) \widehat{I}(a, \chi_{\varepsilon_1,it_1} \times \chi_{\varepsilon_2,it_2}) dt_1 dt_2.$$

with

$$\widehat{I}(a, \chi_{\varepsilon_1,it_1} \times \chi_{\varepsilon_2,it_2}) := \chi_{\varepsilon_1,it_1}^{-1}(a_1) \chi_{\varepsilon_2,it_2}^{-1}(a_2).$$

Proof. This follows from Lemmas 4.1 and 4.4 easily. □

Proposition 4.7. *Let $h = zk_\theta$ with $z > 0$ and $\theta \not\equiv 0, \pi \pmod{2\pi\mathbb{Z}}$. Then,*

$$(4.5) \quad \begin{aligned} I^G(h; f) &= \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{l=1}^{\infty} \widehat{f}(\sigma_{l,it}) \widehat{I}(zk_\theta, \sigma_{l,it}) dt \\ &+ \left(\frac{1}{2\pi}\right)^2 \iint_{\mathbb{R}^2} \frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 \in \{0,1\}} \widehat{f}(\chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2}) \widehat{I}(zk_\theta, \chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2}) dt_1 dt_2 \end{aligned}$$

with

$$\begin{aligned} \widehat{I}(zk_\theta, \sigma_{l,it}) &:= 2 \sin |\theta| z^{-it} \frac{e^{-i\theta} + e^{i\theta}}{e^{-i\theta} - e^{i\theta}}, \\ \widehat{I}(zk_\theta, \chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2}) &:= 2(\chi_{\varepsilon_1, it_1} \chi_{\varepsilon_2, it_2})^{-1}(z) \frac{e^{(|\theta| - \pi/2)(t_1 - t_2)} + (-1)^{\varepsilon_1 + \varepsilon_2} e^{-(|\theta| - \pi/2)(t_1 - t_2)}}{e^{(t_1 - t_2)\pi/2} + (-1)^{\varepsilon_1 + \varepsilon_2} e^{-(t_1 - t_2)\pi/2}} \end{aligned}$$

Proof. . □

The set $\Pi_u(\mathbb{R}^\times) = \{\chi_{\varepsilon, it} \mid \varepsilon \in \{0, 1\}, t \in \mathbb{R}\}$ is endowed a measure defined by

$$\int_{\Pi_u(\mathbb{R}^\times)} \phi(\chi) d\chi := \frac{1}{2\pi} \int_{\mathbb{R}} \frac{1}{2} \sum_{\varepsilon \in \{0,1\}} \phi(\chi_{\varepsilon, it}) dt$$

The set

$$\Pi_t(G_2) := \{\chi_{\varepsilon_1, it} \times \chi_{\varepsilon_2, it_2} \mid \varepsilon_1, \varepsilon_2 \in \{0, 1\}, (t_1, t_2) \in \mathbb{R}^2\} \bigsqcup \{\sigma_{l, it} \mid l \in \mathbb{Z}_{>0}, t \in \mathbb{R}\}$$

has a natural topology and is endowed with the measure defined by

$$\begin{aligned} \int_{\Pi_t(G_2)} \phi(\pi) d\pi &:= \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{l=1}^{\infty} \phi(\sigma_{l, it}) dt \\ &+ \left(\frac{1}{2\pi}\right)^2 \iint_{\mathbb{R}^2} \frac{1}{4} \sum_{\varepsilon_1, \varepsilon_2 \in \{0,1\}} \phi(\chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2}) dt_1 dt_2 \end{aligned}$$

Then, formulae (4.4) and (4.5) are written in a uniform way:

$$I(\gamma, f) = \int_{\Pi_t(G_2)} \widehat{f}(\pi) \widehat{I}(\gamma, \pi) d\pi, \quad \gamma \in \{a, zk_\theta\},$$

where $\widehat{I}(\gamma, \pi) = 0$ if $\gamma = a$ and π is essentially square-integrable representation.

4.6. Tempered representations. For any $p, q \in \mathbb{Z}_{\geq 0}$ such that $2q + p = n$, let $\Pi_t^{(q,p)}(G_n)$ be the (multi-)set of all irreducible unitary representations of the form

$$(4.6) \quad \pi = (\pi_1 \boxtimes \cdots \boxtimes \pi_q \boxtimes \chi_1 \boxtimes \cdots \boxtimes \chi_p)^{G_n}, \quad (\pi_j \in \Pi_t(G_2), \chi_k \in \Pi_u(\mathbb{R}^\times)).$$

By identifying this multi-set with the product set $\Pi_t(G_2)^q \times \Pi_u(\mathbb{R}^\times)^p$, we endow $\Pi_t^{(q,p)}(G_n)$ with the product topology and the product measure. Then we form the disjoint union of all $\Pi_t^{(q,p)}(G_n)$ with $(q, p) \in \mathbb{Z}_{\geq 0}^2$ such that $2q + p = n$, giving the union the unique topology such that each $\Pi_t^{(q,p)}(G_n)$ is an open subspace. An irreducible unitary representation of G is called to be tempered if its global character $I_G(\pi)$ extends to a continuous linear functional on Harish-Chadra's Schwartz space $\mathcal{C}(G)$. For G_n , the unitary dual $\Pi_u(G_n)$ is completely determined by Vogan and Tadic([17]); from that, it is known that any tempered representation is equivalent to one of the representations in (4.6).

We obtain an rigorous replacement of the heuristic formula (4.1):

Corollary 4.8. *If*

$$\gamma = \text{diag}(z_1 k_{\theta_1}, \dots, z_q k_{\theta_q}, a_1, \dots, a_p) \in G_n, \quad 2q + p = n$$

is a regular semi-simple element with p real eigen values $a_1, \dots, a_p$ and q non-real eigenvalues $z_j e^{\sqrt{-1}\theta_j}$. Then, the normalized orbital integral (Definition 8) of f along the G -orbit of γ is written as a superposition of tempered characters as in

$$I^G(\gamma; f) = \int_{\Pi_t(G_n)} \widehat{f}(\pi) \widehat{I}(\gamma, \pi) d\pi,$$

where, for π as in (4.6), we set

$$\widehat{I}(\gamma, \pi) = \prod_{j=1}^q \widehat{I^{G_2}}(z_j k_{\theta_j}, \pi_j) \prod_{k=1}^p \chi_k(a_k)$$

Proof. This is proved by the descent formula (Lemma 4.1) and Lemma 4.4 and Lemmas 4.6 and 4.7. $\square$

Tempered representations are important in the theory of automorphic forms:

- The Selberg-Petersson conjecture asserts that the Laplace eigen values of Hecke-Maass cusp forms $f(\tau)$ on the upper-half plane on a congruence subgroup $\Gamma_0(N)$ are no less than $1/4$. This conjecture is equivalent to saying that the automorphic form on G_2 corresponding to $f(\tau)$ generates a tempered representation of $\text{GL}_2(\mathbb{R})$.
- It is a theorem by Wallach that if a square-integrable form $\phi \in \mathcal{A}(\Gamma; \tau, \chi)$ is contained in a closed irreducible subspace V of $L^2(\Gamma \backslash G^1)$ and V is a tempered representation of G , then it is a cusp form.

4.7. Orbital integrals for general elements. By the theory of Jordan canonical form, any element of $G = G_n$ is $\text{GL}_n(\mathbb{C})$ -conjugate to a matrix of Jordan canonical form

$$\gamma = \bigoplus_{j=1}^n J_{\lambda_j, \ell_j},$$

where $\lambda_1, \dots, \lambda_n$ is the sequence of eigenvalues of γ counted with multiplicities and $J_{\lambda, \ell}$ denotes the Jordan cell of eigen value $\lambda \in \mathbb{C}^\times$ of degree ℓ (with $J_{\lambda, 1}$ representing the 1×1 matrix (λ)). For simplicity, we assume that $\lambda_j \in \mathbb{R}^\times$ for all j . And let $a_1 > a_2 > \dots > a_q$ denote different eigenvalues of γ and m_j the multiplicity of a_j . Then, the centralizer of G_γ is contained in the Levi subgroup $M_{m_1, \dots, m_q}$, so that Lemma 4.1 can be applied to the element and $P = P_{m_1, \dots, m_q}$. Through the map $f \mapsto f_P$, the descent formula reduces the study of orbital integrals $I^G(\gamma; f)$ to the study of orbital integrals $I^{G_{m_j}}(\delta; \cdot)$ on general linear groups of smaller size, with $\delta \in G_{m_j}$ a block diagonal matrix $\text{diag}(J_{a_j, k_1}, \dots, J_{a_j, k_t})$ of Jordan cells of the same eigenvalue a_j . The orbital integrals of this kind is called the unipotent orbital integrals, which are ingredients of the geometric side of the Arthur-Selberg trace formula of non-compact quotients $\Gamma \backslash G^1$. By Harish-Chandra, it is known in theory, that any unipotent orbital integral is written as a limit of derivatives of regular semisimple orbital integrals. The easiest case is when the number of the Jordan cell is 1, i.e, $\gamma = J_{\lambda, n}$. If the eigenvalue $\lambda = 1$, the element $u_{\text{reg}} = J_{1, n}$ is called the regular unipotent element. It is easy to show the limit formula

$$\text{Orb}(u_{\text{reg}}, f) = \lim_{a \rightarrow 1} \text{Orb}([a], f),$$

where $[a] = \text{diag}(a^{n-1}, a^{n-2}, \dots, a, 1)$ for $a \in \mathbb{R}^\times$.

Another extreme case is when all the Jordan cells are size 1, i.e, δ is an element of the center; this case is important in the derivation of the Plancherel formula. For $G = G_2$, it holds that

$$2\pi f(z) = \lim_{\theta \rightarrow +0} \frac{d}{d\theta} I^{G_2}(zk_\theta, f), \quad f \in C_c^\infty(G_2)$$

([12, Chap XI §3, Formula (11.33b)]) By using this, from Proposition 4.7, we get the following:

Proposition 4.9 (Plancherel formula for $\text{GL}_2(\mathbb{R})$). *Let $f \in C_c^\infty(G_2)$. For $z \in \mathbb{R}^\times$,*

$$\begin{aligned} f(z1_2) &= \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{l=1}^{\infty} l \widehat{f}(\sigma_{l,it}) z^{-it} dt \\ &+ \frac{1}{8\pi^2} \sum_{\varepsilon_1, \varepsilon_2 \in \{0,1\}} \text{sgn}^{\varepsilon_1 + \varepsilon_2}(z) \iint_{\mathbb{R}^2} \widehat{f}(\chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2}) \\ &\times z^{-i(t_1+t_2)} (t_1 - t_2) \frac{e^{-\pi(t_1-t_2)/2} + (-1)^{\varepsilon_1 + \varepsilon_2} e^{\pi(t_1-t_2)/2}}{e^{\pi(t_1-t_2)/2} + (-1)^{\varepsilon_1 + \varepsilon_2} e^{-\pi(t_1-t_2)/2}} dt_1 dt_2 \end{aligned}$$

For $G = G_3$, it holds that

$$32\pi^2 f(z) = \lim_{\theta \rightarrow 0} \lim_{t \rightarrow 0} \left(9 \frac{\partial^2}{\partial t^2} \frac{\partial}{\partial \theta} + \frac{\partial^3}{\partial \theta^3} \right) I^{G_3} \left(\begin{bmatrix} e^{-t/3} k_\theta & 0 \\ 0 & e^{2t/3} \end{bmatrix}; f \right), \quad f \in C_c^\infty(G_3)$$

([12, Proposition 11.17(p.420)])³ By this, using Corollary 4.8, we obtain the following.

Proposition 4.10 (Plancherel formula for $\text{GL}_3(\mathbb{R})$). *For $f \in C_c^\infty(G_3)$,*

$$\begin{aligned} f(1_3) &= \frac{1}{8} \sum_{\varepsilon_1, \varepsilon_2, \varepsilon_3 \in \{0,1\}} \left(\frac{1}{2\pi} \right)^3 \int_{\mathbb{R}^3} \frac{1}{8\pi^2} (t_1 - t_2)(t_2 - t_3)(t_3 - t_1) \frac{e^{-\pi(t_1-t_2)/2} - (-1)^{\varepsilon_1 + \varepsilon_2} e^{\pi(t_1-t_2)/2}}{e^{-\pi(t_1-t_2)/2} + (-1)^{\varepsilon_1 + \varepsilon_2} e^{\pi(t_1-t_2)/2}} \\ &\times \widehat{f}(\chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2} \times \chi_{\varepsilon_3, it_3}) dt_1 dt_2 dt_3 \\ &+ \frac{1}{2} \sum_{\varepsilon \in \{0,1\}} \left(\frac{1}{2\pi} \right)^2 \int_{\mathbb{R}^2} \sum_{l=1}^{\infty} \frac{l}{16\pi^2} \{ (t_1 - 2t_2)^2 + l^2 \} \widehat{f}(\sigma_{l, it_1} \chi_{\varepsilon, it_2}) dt_1 dt_2. \end{aligned}$$

5. PROOF OF THE EXPLICIT TRACE FORMULA

5.1. Harmonic analysis on GL_3 . For quasi-characters χ_1, χ_2, χ_3 of $\mathbb{R}^\times$, we have defined a continuous Hilbert representation $\chi_1 \times \chi_2 \times \chi_3 := (\chi_1 \boxtimes \chi_2 \boxtimes \chi_3)^{G_3}$ on a Hilbert space $\mathcal{K}(\chi_1, \chi_2, \chi_3)$ of all measurable functions $f : G_3 \rightarrow \mathbb{C}$ that satisfies

$$f\left(\begin{bmatrix} t_1 & * & * \\ 0 & t_2 & * \\ 0 & 0 & t_3 \end{bmatrix} g\right) = \prod_{j=1}^3 \chi_j(t_j) |t_1| |t_3|^{-1} f(g), \quad \forall \begin{bmatrix} t_1 & * & * \\ 0 & t_2 & * \\ 0 & 0 & t_3 \end{bmatrix} \in P_{1,1,1}$$

for almost all $g \in G_3$. For $\chi_1, \chi_2, \chi_3 \in \Pi_u(\mathbb{R}^\times)$, it is known that the representation $\chi_1 \times \chi_2 \times \chi_3$ is unitary irreducible.

³The coefficient $32\pi^2$ of $f(1_3)$ is determined by comparing the result with the spherical Plancherel formula in [?Helgason]

Let (σ, V) be a continuous representation on a Hilbert space of G_2 and χ a quasi-character of $\mathbb{R}^\times$. We have defined a continuous Hilbert representation $\sigma \times \chi$ on a Hilbert space $\mathcal{K}(\sigma, \chi)$ of all measurable functions $f : G_3 \rightarrow V$ such that

$$f\left(\begin{bmatrix} m_1 & * \\ 0 & t_2 \end{bmatrix} g\right) = |\det(m_1)t_2^{-1}| \chi(t_2) \sigma(m_1)(f(g)), \quad \forall \begin{bmatrix} m_1 & * \\ 0 & t_2 \end{bmatrix} \in P_{2,1}.$$

It is known that when σ is unitary irreducible and χ is unitary, then $\sigma \times \chi$ is irreducible and unitary. For $\chi \in \Pi_u(\mathbb{R}^\times)$, let $\mathbf{1}_{G_3}[\chi]$ be the 1 dimensional representation of $G_3 = Z_3 \mathrm{SL}_2(\mathbb{R})$ defined by $(z1_3)h \mapsto \chi(z_1)$ with $z_1 \in \mathbb{R}^\times$ and $h \in \mathrm{SL}_3(\mathbb{R})$. Any 1-dimensional representation is obtained this way. We quote the main result of [17] specialized to G_3 .

Theorem 5.1. *The set $\Pi_u(G_3)$ is exhausted by the following representations:*

(I) *The minimal principal series representations*

$$\chi_1 \times \chi_2 \times \chi_3 \quad (\chi_1, \chi_2, \chi_3 \in \Pi_u(G_3))$$

(II) *The $P_{2,1}$ -principal series representations:*

$$\sigma_{l,it} \times \chi, \quad (\chi \in \Pi_u(\mathbb{R}^\times), l \in \mathbb{Z}_{\geq 0}, t \in \mathbb{R}).$$

(III) *The non-tempered $P_{2,1}$ -principal series:*

$$\mathbf{1}_{G_2}^\varepsilon[\chi_1] \times \chi_2, \quad (\chi_1, \chi_2 \in \Pi_u(\mathbb{R}^\times)).$$

(IV) *The complementary representations:*

$$\chi_{-t+i\lambda_1, \varepsilon_1} \times \chi_{t+i\lambda_1 \varepsilon_1} \times \chi_{i\lambda_2, \varepsilon_2} \quad (t, \lambda_1, \lambda_2 \in \mathbb{R}, \varepsilon_1, \varepsilon_2 \in \{0, 1\}).$$

(V) *The one dimensional representations:*

$$\mathbf{1}_{G_3}[\chi], \quad (\chi \in \Pi_u(\mathbb{R}^\times), \chi(-1) = 1)$$

defined by $zg \mapsto \chi(z)$ for $z \in \mathbb{R}^\times, g \in G_3^1$.

The infinitesimal characters $(H_1, H_2, H_3) \pmod{S_3}$ of these representations π are given as follows:

	π	$(H_1, H_2, H_3) \pmod{S_3}$
(I)	$\chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2} \times \chi_{\varepsilon_3, it_3}$	(it_1, it_2, it_3)
(II)	$\sigma_{l, it} \times \chi_{\varepsilon, it'}$	$(\frac{l+it}{2}, \frac{-l+it}{2}, it')$
(III)	$\mathbf{1}_{G_2}[\chi_{\varepsilon_1, it_1}] \times \chi_{\varepsilon_2, it_2}$	$(\frac{1+it_1}{2}, \frac{-1+it_1}{2}, it_2)$
(IV)	$\chi_{\varepsilon_1, -t+i\lambda_1} \times \chi_{\varepsilon_1, t+i\lambda_1} \times \chi_{\varepsilon_2, i\lambda_2}$	$(-t+i\lambda_1, t+i\lambda_1, i\lambda_2)$
(V)	$\mathbf{1}_{G_3}[\chi_{\varepsilon, it}]$	$(1 + \frac{it}{3}, \frac{it}{3}, -1 + \frac{it}{3})$

Remark 5 (The Langlands L -factor). *Let $L(s, \pi)$ denote the standard L -factor of a unitary representation $\pi \in \Pi_u(G_3)$, which is defined to be the Artin-Deligne-Langlands L -factor of the Langlands parameter of π . Then,*

$$L(s, \pi) = \Gamma_{\mathbb{R}}(s + H_1) \Gamma_{\mathbb{R}}(s + H_2) \Gamma_{\mathbb{R}}(s + H_3)$$

except for case (II), in which case it is

$$L(s, \pi) = \Gamma_{\mathbb{C}}\left(\frac{l+it}{2} + s\right) \Gamma_{\mathbb{R}}(s + it').$$

The tempered representations are the union of type (I) and (II) representations:

$$\begin{aligned} \Pi_t(G_3) = & \{ \chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2} \times \chi_{\varepsilon_3, it_3} \mid (\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{0, 1\}^3, (t_1, t_2, t_3) \in \mathbb{R} \} \\ & \bigsqcup \{ \sigma_{l, it_1} \times \chi_{\varepsilon, it_2} \mid l \in \mathbb{Z}_{\geq 1}, t_1, t_2 \in \mathbb{R}, \varepsilon \equiv l+1 \pmod{2} \}. \end{aligned}$$

Note that

$$\sigma_{0,it} \times \chi = \chi_{1,it/2} \times \chi_{0,it/2} \times \chi.$$

The set $\Pi_t(G_3)$ is in a bijective correspondence with the space of parameters $((\varepsilon_1, t_1), (\varepsilon_2, t_2), (\varepsilon_3, t_3))$ and $((l, t_1, t_2):$

$$(5.1) \quad \left((\{0, 1\} \times \mathbb{R})^3 / S_3 \right) \sqcup \left((\mathbb{Z}_{>0} \times \{0, 1\} \times \mathbb{R}^2) \right)$$

where S_3 acts on the triples of pairs by permutation of the pairs.

For $m \in \mathbb{Z}_{\geq 0}$, let $\tau_m^\pm$ be the unique representation of $O(3) = \{\pm 1_3\} \times \text{SO}(3)$ such that $\tau_m^\pm(\pm 1_3) = \pm 1$ and $\dim(\tau_m^\pm) = 2m + 1$. By the theory of highest weight, it is proved that $\Pi_u(O(3)) = \{\tau_m^+, \tau_m^- \mid m \in \mathbb{Z}_{\geq 0}\}$. Since $\pi \in \Pi_u(G_3)$ is isomorphic to the parabolically induced representation $\text{Ind}_P^G(\sigma)$ from an irreducible representation σ of the Levi subgroup M of a standard parabolic P , we can calculate $[\pi; \tau]$ for $\tau \in \Pi_u(O(3))$ by the Frobenius reciprocity,

$$[\text{Ind}_P^G(\sigma) : \tau] = \dim \text{Hom}_{M \cap O(3)}(\sigma, \tau|_{M \cap O(3)}),$$

reducing the calculation to the irreducible decomposition of the restriction $\tau|_{M \cap O(3)}$, which is easier. Thus, we can prove the following lemmas:

Lemma 5.2 (Spherical principal series). *The trivial representation τ_0^+ of $O(3)$ occurs in $\pi \in \Pi_t(G_3)$ if and only if $\pi \cong \chi_{0,it_1} \times \chi_{0,it_2} \times \chi_{0,it_3}$.*

This lemma means that the condition $[\pi : \tau_0^+] > 0$ separate out $\chi_{0,it_1} \times \chi_{0,it_2} \times \chi_{0,it_3}$ from other representations in $\Pi_t(G_3)$.

Lemma 5.3. *Let $m \in \mathbb{Z}_{>1}$. Then,*

- (i) *A representation $\pi \in \Pi_t(G_3)$ satisfies $[\pi : \tau_m^+] > 0$ if and only if it is isomorphic to*

$$\sigma_{m-1,it_1} \times \chi_{0,t_2}, \quad \chi_{\varepsilon_1,it_1} \times \chi_{\varepsilon_2,it_2} \times \chi_{\varepsilon_3,it_3}$$

with $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ is S_3 -equivalent to $(0, 0, 0)$ or $(1, 1, 0)$. We have

$$(5.2) \quad [\sigma_{m-1,it_1} \times \chi_{0,t_2}; \tau_m^+] = 1.$$

- (ii) *A representation $\pi \in \Pi_t(G_3)$ satisfies $[\pi : \tau_m^-] > 0$ if and only if it is isomorphic to one of the followings:*

$$\sigma_{m-1,it_1} \times \chi_{1,it_2}, \quad \chi_{\varepsilon_1,it_1} \times \chi_{\varepsilon_2,it_2} \times \chi_{\varepsilon_3,it_3}$$

with $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ is S_3 -equivalent to one of $(1, 1, 1), (1, 0, 0)$. We have

$$[\sigma_{m-1,it_1} \times \chi_{0,t_2}; \tau_m^-] = 1.$$

The second lemma means that requiring $[\pi : \tau_m^\pm] > 0$ is not enough to isolate $\sigma_{l,it} \times \chi$ among all $\Pi_t(G_3)$.

5.1.1. Paley-Wiener theorem. The space of the Schwartz functions $\mathcal{S}(\mathbb{R}^p)$ is invariant by the Fourier transform

$$\widehat{f}(\xi) := \int_{\mathbb{R}^p} f(x) e^{2\pi i x \xi} dx.$$

Actually, $f \mapsto \widehat{f}$ is a linear bijection from $\mathcal{S}(\mathbb{R}^p)$ to itself. The classical Paley-Wiener theorem identifies the Fourier image of $C_c^\infty(\mathbb{R}^p)$ with the space $\text{PW}(\mathbb{C}^p)$ of all entire

functions $\phi : \mathbb{C}^p \rightarrow \mathbb{C}$ that satisfies the estimate: there exists $N \in \mathbb{R}$ such that for all $r \in \mathbb{Z}_{>0}$,

$$\sup_{s \in \mathbb{C}^p} (|f(s)| e^{-N\|\text{Im}(s)\|} (1 + \|s\|)^r) < +\infty.$$

When we view the map which sends $f \in C_c^\infty(G_3)$ to the function $\widehat{f} : \Pi_t(G) \rightarrow \mathbb{C}$ defined by

$$\widehat{f}(\pi) := \text{trace}(\pi(f)), \quad \pi \in \Pi_t(G)$$

as an analogue of the Fourier transform, there is a theorem due to Clozel-Delorme ([8]) that identifies the Fourier image of $C_c^\infty(G)$. We recall it for $G = G_3$. We need a complexification of the real space $\Pi_t(G_3)$, which is defined by embedding each $\mathbb{R}^3$ and $\mathbb{R}^2$ in (5.1) by their ambient complex spaces $\mathbb{C}^3$ and $\mathbb{C}^2$, respectively.

Theorem 5.4. *Given a collection of functions*

$$\phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3} : \mathbb{C}^3 \longrightarrow \mathbb{C}, \quad \phi_{l, \varepsilon} : \mathbb{C}^2 \longrightarrow \mathbb{C}$$

for $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{0, 1\}^3$ and $(l, \varepsilon) \in \mathbb{Z}_{>0} \times \{0, 1\}$. *There exists a function $f \in C_c^\infty(G_3)$ such that*

$$\widehat{f}(\chi_{\varepsilon_1, it_1} \times \chi_{\varepsilon_2, it_2} \times \chi_{\varepsilon_3, it_3}) = \phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3}(t_1, t_2, t_2), \quad \widehat{f}(\sigma_{l, it_1} \times \chi_{\varepsilon, it_2}) = \phi_{l, \varepsilon}(t_1, t_2)$$

for all tempered representations if and only if the following conditions are satisfied:

- (i) $\phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3} \in \text{PW}(\mathbb{C}^3)$ and $\phi_l \in \text{PW}(\mathbb{C}^2)$ for all $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{0, 1\}^3$ and $(l, \varepsilon) \in \mathbb{Z}_{>0} \times \{0, 1\}$.
- (ii) We have $\phi_{\varepsilon'_1, \varepsilon'_2, \varepsilon'_3}(s'_1 s'_2, s'_3) = \phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3}(s_1, s_2, s_3)$ if $((\varepsilon'_1, s'_1), (\varepsilon'_2, s_2), (\varepsilon'_3, s_3))$ and $((\varepsilon_1, s_1), (\varepsilon_2, s_2), (\varepsilon_3, s_3))$ are in the same S_3 -orbit.
- (iii) There exists a finite subset $\mathcal{C} \subset \Pi_u(\text{SO}(3))$ such that

$$\phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3}(s_1, s_2, s_3) = 0 \text{ if } [\chi_{\varepsilon_1, 0} \times \chi_{\varepsilon_2, 0} \times \chi_{\varepsilon_3, 0}; \tau] = 0 \ (\forall \tau \in \mathcal{C})$$

and

$$\phi_{l, \varepsilon}(s) = 0 \text{ if } [\sigma_{l, 0} \times \chi_{\varepsilon, 0}; \tau] = 0 \ (\forall \tau \in \mathcal{C})$$

Note that the multiplicity of $\tau \in \Pi_u(\text{SO}(3))$ in $\chi_{\varepsilon_1, s_1} \times \chi_{\varepsilon_2, s_2} \times \chi_{\varepsilon_3, s_3}$ does not depend on the continuous parameter (s_1, s_2, s_3) . The same is remark applied to $\sigma_{l, s_1} \times \chi_{\varepsilon, s_2}$.

Corollary 5.5. *Let $l \in \mathbb{Z}_{>0}$. Then, for any given $\phi \in \text{PW}(\mathbb{C}^2)$, there exists a function $f \in C_c^\infty(G_3)$ such that for $\pi \in \Pi_t(G_3)$,*

$$\widehat{f}(\pi) = 0$$

unless $\pi \cong \sigma_{l, it_1} \times \chi_{0, it_2}$ for some $(t_1, t_2) \in \mathbb{R}^2$, in which case

$$\widehat{f}(\sigma_{l, it_1} \times \chi_{0, it_2}) = \phi(t_1, t_2) \quad \forall (t_1, t_2) \in \mathbb{R}^2$$

Let $f \in C_c^\infty(G_3)$ be as in Corollary 5.5; it tells us nothing about the values $\widehat{f}(\pi)$ for non-tempered $\pi \in \Pi_u(G_3)$ explicitly, but we can determine them from its behavior on tempered representations. For example, we know that the function

$$\widehat{f}(\chi_{\varepsilon_1, s_1} \times \chi_{\varepsilon_1, s_2} \times \chi_{\varepsilon_2, s_3})$$

is an entire function on $\mathbb{C}^3$ and identically zero on the tempered locus $(i\mathbb{R})^3$. Thus, by the analytic continuation, $\widehat{f}(\pi) = 0$ for π in the complementary series, because $f(\pi)$ is a value of the function above at $s_1 = -t + i\lambda_1$, $s_2 = t + i\lambda_1$, $s_3 = i\lambda_2$ with $t, \lambda_1, \lambda_2 \in \mathbb{R}$.

By the exact sequence (4.3), we have an exact sequence of $(\mathfrak{g}_3^{\mathbb{C}}, O(3))$ -module

$$0 \longrightarrow \sigma_{1,it} \times \chi \longrightarrow \chi_{0,(1+it)/2} \times \chi_{0,(-1+it)/2} \times \chi \longrightarrow \mathbf{1}_{G_2}^0[\chi_{0,it}] \times \chi \longrightarrow 0.$$

By a property of the global characters, this yields the equality

$$\widehat{f}(\chi_{0,(1+it)/2} \times \chi_{0,(-1+it)/2} \times \chi) = \widehat{f}(\sigma_{1,it} \times \chi) + \widehat{f}(\mathbf{1}_{G_2}^0[\chi_{0,it}] \times \chi).$$

As we saw above, the left side is zero. The first term in the right side is zero if $l > 1$ and equals the given value $\phi(t_1, t_2)$ if $l = 1$. Thus, we have the value $\widehat{f}$ at a non-tempered representation $\mathbf{1}_{G_2}^0[\chi_{0,it}] \times \chi$.

To see the values at one dimensional representations, especially the trivial representation $\mathbf{1}_{G_3}$ of G_3 , we use the following lemma.

Lemma 5.6. *There is an exact sequence of $(\mathfrak{g}_3^{\mathbb{C}}, O(3))$ -modules:*

$$0 \longrightarrow \sigma_{2,0} \times \chi_{1,0} \longrightarrow \chi_{1,0} \times \chi_{0,0} \times \chi_{-1,0} \longrightarrow \mathbf{1}_{G_3} \longrightarrow 0$$

For any function f as in Corollary 5.5, We have $\widehat{f}(\mathbf{1}_{G_3}) = -\phi(0, 0)$ if $l = 2$ and $\widehat{f}(\mathbf{1}_{G_3}) = 0$ if $l \neq 2$.

Proof. By Langlands classification ([7, §IV], [12, Chap. VIII, §15 Theorem 8.54]), we have a surjective $(\mathfrak{g}_3^{\mathbb{C}}, O(3))$ -homomorphism $J : \chi_{1,0} \times \chi_{0,0} \times \chi_{-1,0} \longrightarrow \mathbf{1}_{G_3}$. Let π be the kernel of J . Then, J should have the same infinitesimal character as $\mathbf{1}_{G_3}$, i.e., $(1, 0, -1) \pmod{S_3}$. The possibilities of irreducible $(\mathfrak{g}_3^{\mathbb{C}}, O(3))$ -modules having this infinitesimal character are $\sigma_{2,0} \times \chi_{1,0}$ and $\mathbf{1}_{G_3}$. As $\mathbf{1}_{G_3}$ being the Langlands quotient of $\chi_{1,0} \times \chi_{0,0} \times \chi_{-1,0}$, it does not occur as a subquotient of J ([7, Proposition 4.3]). Thus the semisimplification of J is of the form $(\sigma_{2,0} \times \chi_{1,0})^{m\oplus}$. It remains to show that the multiplicity m is indeed 1. By the Frobenius reciprocity, we easily calculate $[\sigma_{2,0} \times \chi_{1,0}; \tau_3^+] = 1$ and $[\chi_{1,0} \times \chi_{0,0} \times \chi_{-1,0}; \tau_3^+] = 1$. This means that $m = 1$ and $J \cong \sigma_{2,0} \times \chi_{1,0}$ as desired. $\square$

5.2. A sketch of the proof of Theorem 2.3. Let A be a central division algebra of degree 3 over $\mathbb{Q}$. Let $\Gamma = \Gamma_{\mathcal{O}}$ be the associated uniform lattice of $\mathbf{SL}_3(\mathbb{R})$ associated with an order $\mathcal{O} \subset A$. Let $l \in \mathbb{Z}_{>0}$ and $\varepsilon \in \{0, 1\}$ the parity of $l + 1$. We are interested in the space of automorphic forms

$$\mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}) \quad (t \in \mathbb{R})$$

on $G^1 = \{\pm 1_3\} \mathbf{SL}_3(\mathbb{Z})$, where $\tilde{\Gamma} = (\pm 1_3) \Gamma_A$ and $\Xi_{l,it} : Z(\mathfrak{g}^{\mathbb{C}}) \rightarrow \mathbb{C}$ is the infinitesimal character of

$$\pi_l^+(t) := \sigma_{l,-2it/3} \times \chi_{\varepsilon, 2it/3} \quad (t \in \mathbb{R}, \varepsilon \equiv l + 1 \pmod{2}),$$

explicitly it is given as

$$\Xi_{l,it}(D) = \gamma(D) \left(\frac{l}{2} + \frac{it}{3}, \frac{-l}{2} + \frac{it}{3}, \frac{-2it}{3} \right),$$

where $\gamma : Z(\mathfrak{g}^{\mathbb{C}}) \cong \mathbb{C}[H_1, H_2, H_3]^{S_3}$ the Harish-Chandra isomorphism. By Lemma 5.3, we have

$$[\pi_l^+(t) : \tau_{l+1}^+] = 1.$$

Recall the multiplicity $m_{\Gamma}(\pi_l^+(t))$ in Lemma 3.1. Then,

Lemma 5.7. *We have*

$$m_{\tilde{\Gamma}}(\pi_l^+(t)) = \dim_{\mathbb{C}} \mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}).$$

Proof. By the table in Theorem 5.1, we can check that if $\pi \in \Pi_u(G)$ has the infinitesimal character $\Xi_{l,it}$, then π is $(\mathfrak{g}_3^{\mathbb{C}}, O(3))$ -isomorphic to

- $\pi_l^+(t)$ if $l > 2$ or $t \neq 0$,
- $\pi_2^+(0)$ or $\mathbf{1}_{G_3}$ if $l = 2$ and $t = 0$,
- $\pi_1^+(0)$ or $\mathbf{1}_{G_2} \times \chi_{0,0}$ if $l = 1$ and $t = 0$.

The condition $[\pi : \tau_{l+1}^+] > 0$ eliminate the case $\pi = \mathbf{1}_{G_3}$ ($l = 2, t = 0$), but is not enough to eliminate the possibility $\pi = \mathbf{1}_{G_2} \times \chi_{0,0}$ ($l = 1, t = 0$) because $\mathbf{1}_{G_2} \times \chi_{0,0}$ and $\pi_1^+(0)$ both contain τ_2^+ with multiplicity 1. To eliminate $\pi \cong \mathbf{1}_{G_2} \times \chi_{0,0}$, we invoke the theory of automorphic representations of GL_3 . By the transfer of automorphic representation of $(A \otimes_{\mathbb{Q}} \mathbb{A})^{\times}$ established by Flath ([13]), if $m_{\tilde{\Gamma}}(\mathbf{1}_{G_2} \times \chi_{0,0}) > 0$, then there exists a cuspidal automorphic representation $\otimes_p \Pi_p$ of $GL_3(\mathbb{A})$ such that $\Pi_{\infty} \cong \mathbf{1}_{G_2} \times \chi_{0,0}$. By a theorem of Shalika ([15]), cuspidal automorphic representations of $GL_n(\mathbb{A})$ are globally generic. As such, Π_{∞} should be locally generic, i.e., admits a Whittaker model. However, by [17], it is known that $\mathbf{1}_{G_2} \times \chi_{0,0}$ has no Whittaker model. Thus, we have proved that $m_{\tilde{\Gamma}}(\pi) > 0$, $[\pi; \tau_{l+1}^+] > 0$ and $\Xi_{\pi} = \Xi_{l,it}$ implies $\pi \cong \pi_l^+(t)$. Then, the claim follows from this combined with (5.2) and (3.1). $\square$

The numbers $m_{\tilde{\Gamma}}(\pi_l^+(t))$ are encoded in the left side of (3.2).

Lemma 5.8. *we have $\Gamma \cap Z = \{1_3\}$, where $Z := \{z1_3 \mid z \in \mathbb{R}^{\times}\}$ is the center of G . The set $\Gamma - \{1_3\}$ consists of regular semisimple elements of G .*

Proof. Let $\iota_{\mathbb{R}} : A \otimes_{\mathbb{Q}} \mathbb{R} \rightarrow \text{Mat}_3(\mathbb{R})$ be the $\mathbb{R}$ -algebra isomorphism that is implicit in the definition of $\Gamma = \Gamma_{\mathcal{O}}$, i.e., $\Gamma_{\mathcal{O}} := \iota_{\mathbb{R}}(\{a \in \mathcal{O} \mid \text{Nr}_{A/\mathbb{Q}}(a) = 1\})$. Since $\det \iota_{\mathbb{R}}(a) = \text{Nr}_A(a)$ for $a \in A$, we have $\Gamma \cap Z \subset \text{SL}_2(\mathbb{R}) \cap Z = \{1_3\}$. By Lemma 4.2, $\Gamma - (\Gamma \cap Z)$ consists of regular semisimple elements. $\square$

Let $\hat{h}(s) \in \text{PW}(\mathbb{C})$. Choose $\beta \in C_c^{\infty}(\mathbb{R})$ such that $\beta(0) = 1$, and define

$$\hat{h}_{\beta}(s, c) := \hat{h}(s)\hat{\beta}(c), \quad (s, c) \in \mathbb{C}^2.$$

Then $\hat{h}_{\beta}(s, c) \in \text{PW}(\mathbb{C}^2)$. Then, we put this function in a family of functions $\phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3}(s_1, s_2, s_3) \in \text{PW}(\mathbb{C}^3)$, $\phi_{m, \varepsilon}(s_1, s) \in \text{PW}(\mathbb{C}^2)$ with $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{0, 1\}^3$, $(\varepsilon, m) \in \{0, 1\} \times \mathbb{Z}_{\geq 1}$ by setting

$$\phi_{\varepsilon_1, \varepsilon_2, \varepsilon_3}(s_1, s_2, s_3) = 0, \quad \phi_{m, \varepsilon}\left(\frac{2(-s+c)}{3}, \frac{2s+c}{3}\right) := \delta_{m, l} \hat{h}_{\beta}(s, c)$$

By Theorem 5.4, there exists $f \in C_c^{\infty}(G_3)$ such that

$$\begin{aligned} \widehat{f}(\chi_{\varepsilon_1, s_1} \times \chi_{\varepsilon_2, s_2} \times \chi_{\varepsilon_3, s_3}) &= 0, \\ \widehat{f}(\sigma_{m, \frac{-2it+2ci}{3}} \times \chi_{\varepsilon', \frac{2it+ic}{3}}) &= \delta_{m, l} \delta_{\varepsilon', \varepsilon} \hat{h}_{\beta}(it, ic). \end{aligned}$$

If $m_{\tilde{\Gamma}}(\pi) > 0$, then π occurs as a subrepresentation of $L^2(\tilde{\Gamma} \backslash G_3^1)$, which means π is trivial on $z1_3$ ($z \in \mathbb{R}_{>0}$). Among such $\pi \in \Pi_u(G_3)$, the non-tempered representations are $\mathbf{1}_{G_2}[\chi_{\varepsilon', -2it/3}] \times \chi_{\varepsilon', 2it/3}$ ($t \in \mathbb{R}, \varepsilon' \in \{0, 1\}$) and $\mathbf{1}_{G_3}$. By the same argument as in Lemma 5.7, we see that $m_{\tilde{\Gamma}}(\mathbf{1}_{G_2}[\chi_{\varepsilon', -2it/3}] \times \chi_{\varepsilon', 2it/3}) = 0$. The G_3 -invariant part of $L^2(\tilde{\Gamma} \backslash G^1)$ coincides with the space of constant functions, which is one dimensional, i.e., $m_{\tilde{\Gamma}}(\mathbf{1}_{G_3}) = 1$. By Lemma 5.6, we have $\widehat{f}(\mathbf{1}_{G_3}) = -\delta_{l, 2} \hat{h}(0)$. Apply formula (3.2) to the test function f . Then, the spectral side becomes

$$\sum_{t \in \mathbb{R}} m_{\tilde{\Gamma}}(\pi_l^+(t)) \hat{h}(it) - \delta_{l, 2} \hat{h}(0).$$

By Lemma 5.8, the geometric side becomes

$$\mathrm{vol}(\tilde{\Gamma} \backslash G_3^1) f(1_3) + \sum_{\gamma \in \mathfrak{E}(\Gamma)} \mathrm{vol}(\tilde{\Gamma}_\gamma \backslash (G^1)_\gamma) \mathrm{Orb}(\gamma; f)$$

By the Fourier inversion,

$$\frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\sigma_{l, \frac{-2it+2ci}{3}} \times \chi_{\varepsilon', \frac{2it+ic}{3}}) dc = \hat{h}(it)$$

and

$$\frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\pi | \det |^{ic/3}) dc = 0$$

for any $\pi \in \Pi_t(G_3)$ other than $\sigma_{l,it} \times \chi_{\varepsilon,it'}$. Using these, by Corollary 4.8 applied to $\gamma \in \Gamma - \{1_3\}$, which is regular semisimple by Lemma 5.8, we have

$$\mathrm{Orb}(\gamma; f) = |\mathrm{Disc}(\Phi_\gamma)|^{-1/2} \sin(l|\theta(\gamma)|) h(2 \log r_\gamma).$$

By the normalization of Haar measure (not explained in detail), we have

$$\mathrm{vol}(\Gamma_\gamma \backslash \mathrm{SL}_3(\mathbb{R})_\gamma) = 3\mathcal{L}(\gamma_0).$$

where γ_0 is the primitive element for γ . We use the Plancherel formula (Proposition 4.10) of G_3 for $f(1_3)$ to get

$$f(1_3) = \frac{1}{4\pi^3} \int_{\mathbb{R}} l(t^2 + \frac{1}{4}l^2) \hat{h}(t) dt.$$

This complete the proof of the explicit trace formula for $h \in \mathrm{PW}(\mathbb{C})$. Then, Following [9], we can prove Proposition 2.1, which in particular means the growth of $\dim_{\mathbb{C}} \mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it})$ for $t \in \Gamma_l$ is of order $O(t^3)$.

Hence the sum $\sum_{t \in \Lambda} \dim_{\mathbb{C}} \mathcal{A}(\tilde{\Gamma}; \tau_{l+1}^+, \Xi_{l,it}) \hat{h}(t)$ converges absolutely if $\hat{h}(r) = (1 + |\mathrm{Re}(r)|)^{-3-\delta}$. To extend the class of test functions h in the trace formula from the Paley-Wiener class to the class of functions of this order of decay, we check that (i) the series in γ is also absolutely convergent for such functions, and (ii) the Paley-Wiener functions well approximate such functions. $\square$

6. TRACE FORMULAE OF NON-COMPACT QUOTIENTS FOR GL_2

The set of places of $\mathbb{Q}$ is identified with the union of the set of prime numbers, which represent non-archimedean places, and the symbol ∞ , which represents the archimedean place so that $\mathbb{Q}_\infty = \mathbb{R}$. Let $\mathbb{A}$ denote the ring of adeles of $\mathbb{Q}$ and $\mathbb{A}^\times$ the group of ideles. Set $\mathbb{A}^1 := \{a \in \mathbb{A}^\times \mid |a|_{\mathbb{A}} = 1\}$, so that $\mathbb{A} = \mathbb{R}_{>0} \mathbb{A}^1$ and $\mathbb{Q}^\times \backslash \mathbb{A}^1$ is compact.

6.1. The spectral analysis of GL_2 . (This section is a summary of a Japanese article in [16] by Tsuzuki, M. and Wakatsuki, S..) Let $G = \mathbf{GL}_2$ viewed as an algebraic group over $\mathbb{Q}$. The $\mathbb{Q}_p$ -points $G(\mathbb{Q}_p)$ inherits a topology from the ambient space $\mathrm{Mat}_2(\mathbb{Q}_p)$ to become a locally compact topological group. Set $\mathbf{K}_p := G(\mathbb{Z}_p)$ if $p < \infty$. Then, the adelicization $G(\mathbb{A})$ is the union of the direct products

$$G_{\mathbb{A}}(S) := \prod_{p \in S} G(\mathbb{Q}_p) \times \prod_{p \notin S} \mathbf{K}_p$$

over all finite set of places S containing ∞ , endowed with the inductive limit topology, i.e., a function $f(g)$ on $G(\mathbb{A})$ is continuous if and only if the restrictions $f|_{G_{\mathbb{A}}(S)}$ are all continuous. A number $z \in \mathbb{R}_{>0}$ is identified with a point $\underline{z} \in G(\mathbb{A})$ whose

archimedean component is $z1_2$ and whose non-archimedean components are all 1_2 . Then, we have a direct product decomposition $G(\mathbb{A}) = \mathbb{R}_{>0} G(\mathbb{A})^1$ with

$$G(\mathbb{A})^1 := \{g \in G(\mathbb{A}) \mid |\det(g)|_{\mathbb{A}} = 1\},$$

where $|a|_{\mathbb{A}} = \prod_p |a|_p$ is the modulus of an idele $a = (a_p)_p \in \mathbb{A}^\times$. When identified with its natural diagonal image in $G(\mathbb{A})$, the group $G(\mathbb{Q})$ is a discrete subgroup of $G(\mathbb{A})^1$. The direct products

$$\mathbf{K}_{\mathbf{f}} := \prod_{p < \infty} \mathbf{K}_p, \quad \mathbf{K} := O(2)\mathbf{K}_{\mathbf{f}}$$

are also viewed as compact subgroups of $G(\mathbb{A})^1$. The strong approximation property of SL_2 yields the decomposition $G(\mathbb{A}) = G(\mathbb{Q})G(\mathbb{R})\mathbf{K}_{\mathbf{f}}$, which in turn implies that the map which sends a point

$$[(g_p)_{p < \infty}] \in G(\mathbb{Q}) \backslash G(\mathbb{A}) / \mathbb{R}_{>0} \mathbf{K}_{\mathbf{f}} = G(\mathbb{Q}) \backslash G(\mathbb{A})^1 / \mathbf{K}_{\mathbf{f}},$$

represented by $(g_p)_p \in G(\mathbb{A})$ with $\det(g_\infty) > 0$, to the point of $\mathrm{GL}_2(\mathbb{Z}) \backslash G(\mathbb{R})^1$ represented by g_∞ is a bijection preserving the right actions of $G(\mathbb{R})$ on both spaces. By this, we can fix a Haar measure on $G(\mathbb{A})$ so that $\mathrm{vol}(G(\mathbb{Q}) \backslash G(\mathbb{A})^1) = \mathrm{vol}(\mathrm{GL}_2(\mathbb{Z}) \backslash G(\mathbb{R})^1) = \mathrm{vol}(\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R})) < \infty$, i.e., $G(\mathbb{Q})$ is a lattice of $G(\mathbb{A})^1$. We set

$$[G] := G(\mathbb{Q}) \backslash G(\mathbb{A})^1.$$

This is an fancy analogue of the fundamental set of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathfrak{h}$. We can speak of the space

$$\mathcal{L}^2([G]) := L^2(G(\mathbb{Q})\mathbb{R}_{>0} \backslash G(\mathbb{A}))$$

as a unitary representation of $G(\mathbb{A})$ acting by the right translations. The bijection $[G]/\mathbf{K}_{\mathbf{f}} \cong \mathrm{GL}_2(\mathbb{Z}) \backslash G(\mathbb{R})^1$ allows us to lift any automorphic form ϕ on $\mathrm{GL}_2(\mathbb{Z}) \backslash G(\mathbb{R})^1$ to a right $\mathbf{K}_{\mathbf{f}}$ -invariant function on the discrete quotient $[G]$ of adeles. Indeed, the classical cusp forms are mapped to the space of adelic cusp forms $\mathcal{L}_0^2([G])$, which is a closed $G(\mathbb{A})$ -invariant subspace formed by all $\phi \in \mathcal{L}^2([G])$ such that

$$\int_{N(\mathbb{Q}) \backslash N(\mathbb{A})} \phi(ug) \, du = 0$$

for almost everywhere in $g \in G(\mathbb{A})$, where $N = N_{1,1} = \{ \begin{bmatrix} 1 & * \\ 0 & 1 \end{bmatrix} \}$. As in Theorem 3.1, the unitary representation $\mathcal{L}_0^2([G])$ is decomposed to a discrete direct sum of irreducible unitary representations π of $G(\mathbb{A})$ with finite multiplicities $m_0(\pi)$. Indeed, we have

Theorem 6.1 (The multiplicity one theorem). $m_0(\pi) \leq 1$.

Any irreducible subspace of $\mathcal{L}_0^2([G])$ is called a cuspidal automorphic representation of $G(\mathbb{A})$. Representation theoretically speaking, a more natural object to consider is the space $\mathcal{L}_{\mathrm{disc}}^2([G])$, the closure of the sum of irreducible closed subspaces of $\mathcal{L}^2([G])$. Then $\mathcal{L}_{\mathrm{cont}}^2([G])$ (the continuous spectrum) is defined as the orthogonal complement of the discrete spectrum $\mathcal{L}_{\mathrm{disc}}^2([G])$; $\mathcal{L}_{\mathrm{res}}^2([G])$ (the residue spectrum) is defined as the orthogonal complement of $\mathcal{L}_0^2([G])$ in $\mathcal{L}_{\mathrm{disc}}^2([G])$. Thus, we obtain the orthogonal direct sum decomposition:

$$\mathcal{L}^2([G]) = \mathcal{L}_0^2([G]) \hat{\oplus} \mathcal{L}_{\mathrm{res}}^2([G]) \hat{\oplus} \mathcal{L}_{\mathrm{cont}}^2([G]).$$

The theory of Eisenstein series concretely describe the structure of the last two spaces. Let P denote the parabolic subgroup formed by all upper-triangular matrices

in G . Define a function $H_P : G(\mathbb{A}) \rightarrow \mathbb{R}$ by using the Iwasawa decomposition $G(\mathbb{A}) = P(\mathbb{A})\mathbf{K}$ as

$$H_P \left(\begin{bmatrix} t_1 & x \\ 0 & t_2 \end{bmatrix} k \right) := \frac{1}{2} \log |t_1/t_2|_{\mathbb{A}}.$$

Definition 11 (Principal series representation). *Let $\mathbf{H} := L^2(M(\mathbb{Q})N(\mathbb{A})\mathbb{R}_{>0} \backslash G(\mathbb{A}))$. For $s \in \mathbb{C}$, we define a $G(\mathbb{A})$ -action on $\mathbf{H}$ by*

$$\pi_s(g)\phi](x) := e^{-sH_P(xg)} e^{sH_P(x)} \phi(xg), \quad \phi \in \mathbf{H}.$$

Let $\mathbf{H}^0$ denote the space of $\mathbf{K}$ -finite vectors, i.e., those $\phi \in \mathbf{H}$ such that $\dim_{\mathbb{C}} \langle \pi_s(\mathbf{K})\phi \rangle_{\mathbb{C}} < \infty$. Since the abelian group $M(\mathbb{A})$ normalizes $N(\mathbb{A})$, it acts on $\mathbf{H}$ by left translations. For a pair of idele class characters $\mu_i : \mathbb{A}^{\times}/\mathbb{R}_{>0}\mathbb{Q}^{\times} \rightarrow \mathbb{C}^1$ ($i = 1, 2$), let $\mathbf{H}_{\mu}^0$ denote the space of all $\phi \in \mathbf{H}^0$ such that

$$\phi \left(\begin{bmatrix} t_1 & x \\ 0 & t_2 \end{bmatrix} g \right) = \mu_1(t_1)\mu_2(t_2)|t_1/t_2|_{\mathbb{A}}^{1/2} \phi(g), \quad \begin{bmatrix} t_1 & x \\ 0 & t_2 \end{bmatrix} \in P(\mathbb{A}), g \in G(\mathbb{A}).$$

Then $\mathbf{H}^0$ is a Hilbert direct sum of subspaces $\mathbf{H}_{\mu}^0$ for all $\mu = (\mu_1, \mu_2)$.

Now, for $\phi \in \mathbf{H}_{\mu}^0$ and $s \in \mathbb{C}$, the Eisenstein series is defined by the sum

$$\mathbf{E}(\phi, s; g) := \sum_{\gamma \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} \phi^{(s)}(\gamma g), \quad g \in G(\mathbb{A}),$$

where

$$\phi^{(s)}(g) := e^{sH_P(g)} \phi(g), \quad g \in G(\mathbb{A}).$$

To describe the constant term in the Fourier expansion of $\mathbf{E}(\phi, is)$, we need the integral defined by

$$[\mathbf{M}(s)\phi](g) := e^{-H_P(g)} \int_{N(\mathbb{A})} \phi^{(s)}(w_0 u g) du, \quad \phi \in \mathbf{H}_{\mu}^0, g \in G(\mathbb{A})$$

with $w_0 := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The basic properties of $\mathbf{E}(\phi, s)$ and $\mathbf{M}(s)$ are summarized as follows:

Proposition 6.2. (1) *The series $\mathbf{E}(\phi, s; g)$ and the integral $\mathbf{M}(s)\phi(g)$ converges absolutely and normally in (s, g) on $\text{Re}(s) > 1$ defining a holomorphic functions on $\text{Re}(s) > 1$. Moreover, the resulting function $\mathbf{M}(s)\phi$ on $G(\mathbb{A})$ belongs to the space $\mathbf{H}_{\mu}$, and we obtain a $G(\mathbb{A})$ -intertwining operator⁴*

$$\mathbf{M}(s) : (\pi_s, \mathbf{H}_{\mu}^0) \longrightarrow (\pi_{-s}, \mathbf{H}_{\mu}^0).$$

and

$$\int_{N(\mathbb{Q}) \backslash N(\mathbb{A})} \mathbf{E}(\phi, s; ug) du = \phi^{(s)}(g) + [\mathbf{M}(s)\phi]^{(-s)}(g), \quad g \in G(\mathbb{A}), \phi \in \mathbf{H}_{\mu}^0, \text{Re}(s) > 1.$$

(2) *For $\phi \in \mathbf{H}_{\mu}^0$ and $g \in G(\mathbb{A})$, the holomorphic function $s \mapsto \mathbf{E}(\phi, s; g)$ ($\text{Re}(s) > 1$), together with $s \mapsto \mathbf{M}(s)\phi(g)$ ($\text{Re}(s) > 1$), has a meromorphic continuation to $\mathbb{C}$ holomorphic on the closed half-plane $\text{Re}(s) \geq 0$ except for a possible simple pole at $s = 1$, admitting the functional equation*

$$\mathbf{E}(\phi, s; g) = \mathbf{E}(\mathbf{M}(s)\phi, -s; g)$$

⁴Exactly speaking, the group $G(\mathbb{A})$ does not acts on $\mathbf{H}_{\mu}^0$, because $G(\mathbb{R})$ does not preserve $O(2)$ -finite vectors. The $G(\mathbb{A})$ -action means $(\mathfrak{g}^{\mathbb{C}}, O(2)) \times G(\mathbb{A}_{\mathbf{f}})$ -action

- (3) For $\phi \in \mathbf{H}_\mu$ with $\mu = (\mu_1, \mu_2)$, the function $\mathbf{E}(\phi, s; g)$ of s has a pole at $s = 1$ if and only if $\mu_1 = \mu_2$ and

$$\text{Res}_{s=1} \mathbf{E}(\phi, s; g) = \frac{-6}{\pi} \langle \phi, 1 \rangle_{\mathbf{K}} \mu_1(\det g), \quad g \in G(\mathbb{A}).$$

with $\langle \phi, 1 \rangle_{\mathbf{K}} := \int_{\mathbf{K}} \phi(k) dk$.

As $\mathbf{E}(\phi, s; g)$ is holomorphic on the imaginary axis $\text{Re}(s) = 0$, the following definition makes sense:

Definition 12 (Wave packet of the Eisenstein series). For $\phi \in \mathbf{H}_\mu^0$ and a Paly-Wiener function $\mathbf{a}(s) \in \text{PW}(\mathbb{C})$, define

$$\mathbf{S}^*(\mathbf{a} \otimes \phi)(g) := \frac{1}{8\pi i} \int_{\text{Re}(s)=0} \mathbf{E}(\phi, s; g) \mathbf{a}(s) ds, \quad g \in G(\mathbb{A}).$$

Proposition 6.3. For $\phi \in \mathbf{H}_\mu^0$ and $\mathbf{a} \in \text{PW}(\mathbb{C})$, the function $\mathbf{S}^*(\mathbf{a} \otimes \phi)$ belongs to $\mathcal{L}_{\text{cont}}^2([G])$. The subspace $\sum_{\mu} \{\mathbf{S}^*(\mathbf{a} \otimes \phi) \mid \mathbf{a} \in \text{PW}(\mathbb{C}), \phi \in \mathbf{H}_\mu^0\}$ is a dense subspace of $\mathcal{L}_{\text{cont}}^2([G])$. For an idele class character $\eta : \mathbb{A}^\times / \mathbb{R}_{>0} \mathbb{Q}^\times \rightarrow \mathbb{C}^\times$, define

$$\varphi_\eta(g) := \text{vol}(G(\mathbb{Q}) \backslash G(\mathbb{A})^1)^{-1} \eta(\det g), \quad g \in G(\mathbb{A}).$$

Then, φ_η is a unit vector of $\mathcal{L}^2([G])$ and $\mathcal{L}_{\text{res}}^2([G])$ coincides with the $\mathbb{C}$ -span of functions φ_η .

By the residue theorem and Proposition 6.2(3), for $c > 1$,

$$\frac{1}{8\pi i} \int_{\text{Re}(s)=c} \mathbf{E}(\phi, s; g) \mathbf{a}(s) ds = \mathbf{S}^*(\mathbf{a} \otimes \phi)(g) + \delta_{\mu_1, \mu_2} \frac{-1}{4} \text{Res}_{s=1} \mathbf{E}(\phi, s; g)$$

This explains the separation of cases in the following definition. Let $\mu = (\mu_1, \mu_2)$, define a closed $G(\mathbb{A})$ -invariant subspace $\mathcal{L}^2(P, [\mu])$ as follows:

- If $\mu_1 \neq \mu_2$, then $\mathcal{L}^2(P, [\mu])$ is the closure of wave-packets $\mathbf{S}^*(\mathbf{a} \otimes \phi)$ ($\mathbf{a} \in \text{PW}(\mathbb{C})$, $\phi \in \mathbf{H}_\mu^0$) in $\mathcal{L}^2([G])$.
- If $\mu_1 = \mu_2$, then $\mathcal{L}^2(P, [\mu])$ is the sum of the closure of wave-packets $\mathbf{S}^*(\mathbf{a} \otimes \phi)$ ($\mathbf{a} \in \text{PW}(\mathbb{C})$, $\phi \in \mathbf{H}_\mu^0$) and the 1-dimensional space $\mathbb{C}\varphi_{\mu_1}$.

Then, the total L^2 space $\mathcal{L}^2([G])$ is decomposed to an orthogonal direct sum of $G(\mathbb{A})$ -invariant closed subspaces

$$\mathcal{L}^2([G]) = \mathcal{L}_0^2([G]) \hat{\oplus} \widehat{\bigoplus_{[\mu] \in \mathfrak{X}^M} \mathcal{L}^2(P, [\mu])}$$

where $\mathfrak{X}^M$ is the set of all different cosets $[\mu] := \{\mu, \check{\mu}\}$.

6.2. The modified kernel. As in the real case, any function $f \in C_c^\infty(G(\mathbb{A}))$ induces a bounded linear operator $R_G(f)$ on $\mathcal{L}^2([G])$ with the kernel function

$$K^f(g, h) := \sum_{\gamma \in G(\mathbb{Q})} f(g^{-1}\gamma h), \quad (g, h) \in G(\mathbb{A})^1.$$

This is a continuous function on $[G] \times [G]$ so that we may consider its diagonal restriction $k(f; g) := K^f(g, g)$. Our concern is the trace of the operator $R_{G,0}(f) := R_G|_{\mathcal{L}_0^2([G])}$, which is known to be of trace class. To compute the trace, Arthur introduced the modified kernel as

$$k^T(f; g) := K^f(g, g) - \sum_{\delta \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} K_P^f(\delta g, \delta g) \hat{\tau}(H_P(\delta g) - T), \quad g \in G(\mathbb{A})^1$$

depending on a parameter $T > 0$, where $K_P^f(g, h)$ is the kernel operator of the right regular action $R_P(f) = \pi_0(f)$ on $\mathbf{H} = \mathcal{L}^2(M(\mathbb{Q})N(\mathbb{A})\mathbb{R}_{>0} \backslash G(\mathbb{A}))$, or explicitly

$$K_P^f(g, h) := \sum_{\gamma \in M(\mathbb{Q})} \int_{N(\mathbb{A})} f(g^{-1}\gamma uh) du$$

and $\hat{\tau}$ is the characteristic function of $[0, +\infty)$ on $\mathbb{R}$ and $H_P(g) := 2^{-1} \log \delta_P(g)$. Then,

Lemma 6.4. *For sufficiently large $T > 0$, the function $k^T(f; g)$ is integrable on $[G]$.*

The modified trace of $R_G(f)$ is defined by the integral

$$J^T(f) := \int_{[G]} k^T(f; g) dg.$$

Let $\mathfrak{X}$ be the set of pairs (G, π) or $(P, [\mu])$ where π is an irreducible cuspidal automorphic representation of $G(\mathbb{A})^1$ and $[\mu]$ the equivalence classes of pair idele class characters of $\mathbb{A}^\times$. Let $\chi \in \mathfrak{X}$. If $\chi = (P, [\mu])$, let $K_{G,\chi}^f(g, h)$ denote the kernel function of the restriction of the operator $R_G(f)$ to $\mathcal{L}^2(P, [\mu])$ and $K_{P,\chi}^f(g, h)$ the kernel function of the operator $\pi_0(f)$ restricted to the closure of $\mathbf{H}_\mu^0 + \mathbf{H}_{\bar{\mu}}^0$ in the Hilbert space $\mathbf{H}$.

If $\chi = (G, \pi)$, then we set $K_{G,\chi}(g, h)$ to be the kernel function of $R_G(f)$ restricted to $\mathcal{L}^2([G])_\pi$, the π -isotypic component of $\mathcal{L}_0^2([G])$ and $K_{P,\chi}^f(g, h) = 0$.

Then, define

$$k_\chi^T(f; g) := K_{G,\chi}^f(g, g) - \sum_{\gamma \in P(\mathbb{Q}) \backslash G(\mathbb{Q})} K_{P,\chi}^f(\delta g, \delta g) \hat{\tau}(H_P(\delta g) - T).$$

Lemma 6.5. *For sufficiently large $T > 0$, we have*

$$\sum_{\chi \in \mathfrak{X}} \int_{[G]} |k_\chi^T(f; g)| dg < \infty$$

and

$$J^T(f) = \sum_{\chi \in \mathfrak{X}} J_\chi^T(f)$$

with

$$J_\chi^T(f) := \int_{[G]} k_\chi^T(f; g) dg.$$

Let $\mathfrak{X}_G$ denote the set of all $\chi = (G, \pi)$. Then, for $\chi \in \mathfrak{X}_G$, the function $k_\chi^T(f; g)$ is independent of T and we retrieve the desired trace by the formula

$$(6.1) \quad \text{trace}(R_0(f)) = \sum_{\chi \in \mathfrak{X}_G} J_\chi^T(f) = J^T(f) - \sum_{\chi \in \mathfrak{X} - \mathfrak{X}_G} J_\chi^T(f).$$

6.3. The spectral side. The integrals $J_\chi^T(f)$ for $\chi \in \mathfrak{X} - \mathfrak{X}_G$ are explicitly calculated as follows.

Proposition 6.6. *Let $\chi = (P, [\mu])$ with $\mu = (\mu_1, \mu_2)$. If $\mu_1 \neq \mu_2$, then*

$$J_\chi^T(f) = \frac{-1}{4\pi} \int_{\mathbb{R}} \text{trace}(\mathbf{M}(-is)\mathbf{M}'(is)\pi_{is}(f)|_{\mathbf{H}_\mu \oplus \mathbf{H}_{\bar{\mu}}}) ds + T \mathcal{R}_\chi(f)$$

If $\mu_1 = \mu_2$, then

$$\begin{aligned} J_\chi^T(f) &= \frac{-1}{4\pi} \int_{\mathbb{R}} \text{trace}(\mathbf{M}(-is)\mathbf{M}'(is)\pi_{is}(f)|_{\mathbf{H}_\mu}) \, ds \\ &\quad + \int_{\mathbf{G}(\mathbb{A})^1} \mu_1(\det g)f(g) \, dg + \frac{1}{4} \text{trace}(\mathbf{M}(0)\pi_0(f)|_{\mathbf{H}_\mu}) + T \mathcal{R}_\chi(f) \end{aligned}$$

6.4. The geometric side. We compute the integral $J^T(f)$ by dividing the function $k^T(f; g)$ to subseries according to “the $\mathcal{O}$ -equivalence classes” in $G(\mathbb{Q})$. Here, two elements $\gamma, \gamma' \in G(\mathbb{Q})$ are defined to be $\mathcal{O}$ -equivalent if γ_s and γ'_s are $G(\mathbb{Q})$ -conjugate, where γ_s denotes the semisimple part of the Jordan-Chevalley decomposition of γ . Suppose that γ is regular semisimple, i.e., γ has 2 distinct eigenvalues, so that the centralizer of γ is a torus, then the $\mathcal{O}$ -equivalence class of γ and the $G(\mathbb{Q})$ -conjugacy class of γ are the same. If $\{\gamma\}_{G(\mathbb{Q})} \cap P(\mathbb{Q}) = \emptyset$, we say γ is $\mathbb{Q}$ -elliptic. If $\{\gamma\}_{G(\mathbb{Q})} \cap P(\mathbb{Q}) \neq \emptyset$, we say γ is $\mathbb{Q}$ -hyperbolic. Suppose γ is not regular semisimple; then, by the Jordan canonical form, γ is $G(\mathbb{Q})$ -conjugate to either $z1_2$ or $z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ for some $z \in \mathbb{Q}^\times$. Actually, any $\mathcal{O}$ -equivalence class not containing regular semisimple elements is a union of two conjugacy classes $\mathfrak{o}(z) := \{z1_2\} \sqcup \{z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\}_{G(\mathbb{Q})}$ for some $z \in \mathbb{Q}^\times$.

For each $\mathcal{O}$ -equivalence class $\mathfrak{o} \subset G(\mathbb{Q})$, define

$$k_\mathfrak{o}^T(f; g) := K_\mathfrak{o}^f(g, g) - \sum_{\delta \in P(\mathbb{Q}) \setminus G(\mathbb{Q})} K_{P, \mathfrak{o}}^f(\delta g, \delta g) \hat{\tau}(H_P(\delta g) - T),$$

where

$$\begin{aligned} K_\mathfrak{o}^f(g, h) &:= \sum_{\gamma \in \mathfrak{o}} f(g^{-1}\gamma h), \\ K_{P, \mathfrak{o}}^f(g, h) &:= \sum_{\gamma \in M(\mathbb{Q}) \cap \mathfrak{o}} \int_{N(\mathbb{A})} f(g^{-1}\gamma u h) \, du. \end{aligned}$$

Lemma 6.7. *For sufficiently large $T > 0$, we have*

$$\sum_{\mathfrak{o}} \int_{[G]} |k_\mathfrak{o}^T(f; g)| \, dg < \infty.$$

and

$$(6.2) \quad J^T(f) = \sum_{\mathfrak{o}} J_\mathfrak{o}^T(f), \quad J_\mathfrak{o}^T(f) := \int_{[G]} k_\mathfrak{o}^T(f; g) \, dg.$$

The terms $J_\mathfrak{o}^T(f)$ are explicitly calculated as follows.

Proposition 6.8. *Let $\mathfrak{o} = \{\gamma\}_{G(\mathbb{Q})}$ be the $G(\mathbb{Q})$ -conjugacy class of a $\mathbb{Q}$ -elliptic element γ . Then*

$$J_\mathfrak{o}^T(f) = \text{vol}(G(\mathbb{Q})_\gamma \backslash G(\mathbb{A})_\gamma^1) \int_{G(\mathbb{A})_\gamma \backslash G(\mathbb{A})} f(g^{-1}\gamma g) \, dg$$

Let $\mathfrak{o} = \{\gamma\}_{G(\mathbb{Q})}$ be the $G(\mathbb{Q})$ -conjugacy class of a $\mathbb{Q}$ -hyperbolic element γ . Then,

$$J_\mathfrak{o}^T(f) = \frac{1}{2} \sum_{\delta \in M(\mathbb{Q}) \cap \mathfrak{o}} \int_{\mathbf{K}} \int_{N(\mathbb{A})} f(k^{-1}n^{-1}\delta nk)(-H_P(w_0 n)) \, dn dk + T \mathcal{R}_\mathfrak{o}(f).$$

Let $\mathfrak{o} = \{z1_2\} \cup \{z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\}_{G(\mathbb{Q})}$ with $z \in \mathbb{Q}^\times$ be a unipotent $\mathcal{O}$ -equivalence class. Then,

$$J_{\mathfrak{o}}^T(f) = \text{vol}(G(\mathbb{Q}) \backslash G(\mathbb{A})^1) \left\{ f(z1_2) + \lim_{s \rightarrow 1} \frac{d}{ds} (s-1) \int_{\mathbb{A}^\times} F_z(a) |a|_{\mathbb{A}}^{s-1} da \right\} + T \mathcal{R}_{\mathfrak{o}}(f),$$

where F_z is a Schwartz-Bruhat function on $\mathbb{A}$ defined by

$$F_z(u) := \int_{\mathbf{K}} f(k^{-1} z \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} k) dk, \quad u \in \mathbb{A}.$$

Note that the cardinality of the set $M(\mathbb{Q}) \cap \mathfrak{o}$ for a $\mathbb{Q}$ -hyperbolic $\mathfrak{o}$ is 2; indeed, if $\gamma = \text{diag}(a_1, a_2)$ ($a_1 \neq a_2$), then $M(\mathbb{Q}) \cap \mathfrak{o} = \{\text{daig}(a_1, a_2), \text{diag}(a_2, a_1)\}$.

6.5. Arthur-Selberg trace formula. Let $f \in C_c^\infty(G(\mathbb{A}))$. The formulas in Propositions ?? and (6.8) shows that $J_\chi^T(f)$ and $J_{\mathfrak{o}}^T(f)$ are actually polynomials in T of degree at most 1, so that we can speak of their constant terms $J_\chi(f) := J_\chi^0(f)$ and $J_{\mathfrak{o}}(f) := J_{\mathfrak{o}}^0(f)$. By formula (6.2) with formulas in Proposition 6.8 being substituted and by formulae in Propositions 6.6, from (6.1), we obtain

$$(6.3) \quad \text{trace} R_0(f) = \sum_{\mathfrak{o}} J_{\mathfrak{o}}(f) - \sum_{\chi \in \mathfrak{X} - \mathfrak{X}_G} J_\chi(f) + T \mathcal{R}(f).$$

with some linear form $\mathcal{R}(f)$. Since the left side of the equality is independent of T , the coefficient $\mathcal{R}(f)$ of T in the right side should be 0 identically. Thus,

Arthur-Selberg trace formula

$$(6.4) \quad \sum_{\chi \in \mathfrak{X}} J_\chi(f) = \sum_{\mathfrak{o}} J_{\mathfrak{o}}(f),$$

or explicitly

$$\begin{aligned} \text{trace} R_0(f) = & \sum_{\mathfrak{o}(\gamma): \mathbb{Q}\text{-elliptic}} \text{vol}(G(\mathbb{Q})_\gamma \backslash G(\mathbb{A})_\gamma^1) \int_{G(\mathbb{A})_\gamma \backslash G(\mathbb{A})} f(g^{-1} \gamma g) dg \\ & + \sum_{\mathfrak{o}(\gamma): \mathbb{Q}\text{-hyperbolic}} \frac{1}{2} \sum_{\delta \in M(\mathbb{Q}) \cap \mathfrak{o}(\gamma)} \int_{\mathbf{K}} \int_{N(\mathbb{A})} f(k^{-1} n^{-1} \delta n k) (-H_P(w_0 n)) dndk \\ & + \sum_{z \in \mathbb{Q}^\times} \text{vol}(G(\mathbb{Q}) \backslash G(\mathbb{A})^1) \left\{ f(z1_2) + \lim_{s \rightarrow 1} \frac{d}{ds} (s-1) \int_{\mathbb{A}^\times} F_z(a) |a|_{\mathbb{A}}^{s-1} da \right\} \\ & + \sum_{\mu_1 \neq \mu_2} \frac{1}{4\pi} \int_{\mathbb{R}} \text{trace} \left(\mathbf{M}(-it) \mathbf{M}'(it) \pi_{it}(f) |_{\mathbf{H}_\mu \oplus \mathbf{H}_{\bar{\mu}}} \right) dt \\ & + \sum_{\mu_1} \left\{ \frac{1}{4\pi} \int_{\mathbb{R}} \text{trace} \left(\mathbf{M}(-it) \mathbf{M}'(it) \pi_{it}(f) |_{\mathbf{H}_{(\mu_1, \mu_1)}} \right) dt \right. \\ & \quad \left. - \int_{G(\mathbb{A})^1} \mu_1(\det g) f(g) dg + \frac{-1}{4} \text{trace}(\mathbf{M}(0) \pi_0(f) |_{\mathbf{H}_{(\mu_1, \mu_1)}}) \right\}. \end{aligned}$$

6.6. The weighted orbital integrals. For application, we usually take a finite set of places S containing ∞ and choose the test function $f \in C_c^\infty(G(\mathbb{A}))$ of the form

$$(6.5) \quad f(g) = \prod_v f_v(g_v), \quad g = (g_v)_v \in G(\mathbb{A})$$

with $f_v \in C_c^\infty(G(\mathbb{Q}_v))$ such that $f_p = \mathbb{1}_{\mathbf{K}_p}$ for $p \notin S$ and f_v is $\mathbf{K}_v \times \mathbf{K}_v$ -finite, i.e., there exists a finite dimensional representation τ_v of $\mathbf{K}_v$ such that $e_\tau *_{\mathbf{K}_v} f_v *_{\mathbf{K}_v} e_\tau = f_v$ with $e_\tau(k_v) := (\dim \tau)^{-1} \text{tr} \tau(k)$.

Then $\mathbb{Q}$ -hyperbolic and unipotent terms in the geometric side are computed further in terms of linear functionals on $C_c^\infty(G(\mathbb{Q}_v))$, called the orbital integrals and the weighted orbital integrals. For $\gamma \in G(\mathbb{Q}_v)$ with the Jordan-Chevalley decomposition $\gamma = \gamma_s \gamma_u$, define

$$D_G(\gamma) := \det((\text{Id} - \text{Ad}(\gamma))|_{\mathfrak{g}/\mathfrak{g}_{\gamma_s}}).$$

The following definition was already made (Definition 8) when $v = \infty$.

Definition 13 (The orbital integral). *For $\gamma \in G(\mathbb{Q}_v)$ and $f_v \in C_c^\infty(G(\mathbb{Q}_v))$, the integral*

$$I_G^G(\gamma, f_v) := |D_G(\gamma)|^{1/2} \int_{G(\mathbb{Q}_v)_\gamma \backslash G(\mathbb{Q}_v)} f_v(g^{-1}\gamma g) dg$$

is absolutely convergent ([14]). It is called the orbital integral of $f_v \in C_c^\infty(G(\mathbb{Q}_v))$.

The weighted orbital integral is introduced in a general context by Arthur([?]), associated with a standard Levi subgroup M and its element $\delta \in M(\mathbb{Q}_v)$. In the current setting, only choice is M , the diagonal subgroup of G . The definition needs steps. Define

$$v_M(g) := \lim_{s \rightarrow 0} \left(\frac{e^{-H_P(g)}}{s} + \frac{e^{-sH_{\bar{P}}(g)}}{-s} \right), \quad g \in G(\mathbb{Q}_v).$$

where $H_P, H_{\bar{P}} : G(\mathbb{A}) \rightarrow \mathbb{R}_{>0}$ are defined by

$$H_P \left(\begin{bmatrix} t_1 & x \\ 0 & t_2 \end{bmatrix} k \right) = H_{\bar{P}} \left(\begin{bmatrix} t_1 & 0 \\ x & t_2 \end{bmatrix} k \right) = \frac{1}{2} \log |t_1/t_2|_{\mathbb{A}}, \quad t_1, t_2 \in \mathbb{Q}_v^\times, x \in \mathbb{Q}_v, k \in \mathbf{K}_v$$

Definition 14 (The weighted orbital integral). *Let $\delta \in M(\mathbb{Q}_v)$. If $G(\mathbb{Q}_v)_\delta \subset M(\mathbb{Q}_v)$, i.e., $\delta = \text{diag}(a_1, a_2)$ with $a_1 \neq a_2$, then*

$$J_M^G(\gamma, f_v) := |D_G(\gamma)|^{1/2} \int_{G(\mathbb{Q}_v)_\gamma \backslash G(\mathbb{Q}_v)} f(g^{-1}\gamma g) v_M(g) dg$$

is defined to be the weighted orbital integral of $f \in C_c^\infty(G(\mathbb{Q}_v))$. If $G(\mathbb{Q}_v)_\delta \not\subset M(\mathbb{Q}_v)$, i.e., $\delta = z1_2$ with $z \in \mathbb{Q}_v^\times$, then

$$(6.6) \quad J_M^G(\delta, f_v) := \lim_{a \rightarrow 1} \left\{ J_M^G \left(\begin{bmatrix} za & 0 \\ 0 & za^{-1} \end{bmatrix}, f_v \right) + r_A^G(a) J_G \left(\begin{bmatrix} za & 0 \\ 0 & za^{-1} \end{bmatrix}, f_v \right) \right\}$$

with $r_M^G(a) := \log |(1 - a^2)(1 - a^{-2})|^{1/2}$ exists (see Lemma 6.9); the limit value is defined to be the weighted orbital integral of f_v .

Let $\delta \in M(\mathbb{Q}_v) - Z(\mathbb{Q}_v)$, Then $G(\mathbb{Q}_v)_\delta = M(\mathbb{Q}_v)$; by the Iwasawa decomposition of $G(\mathbb{Q}_v)$, and by $v_M(uk) = -H_P(w_0u)$ for $k \in \mathbf{K}_v$ and $u \in N(\mathbb{Q}_v)$, we have

$$J_M^G(\delta, f_v) = \int_{\mathbf{K}_v} \int_{N(\mathbb{Q}_v)} f_v(k^{-1}u^{-1}\delta uk) (-H_P(w_0u)) dn dk.$$

Lemma 6.9. *Let $\delta = \text{diag}(za, za^{-1})$ with $z, a \in \mathbb{Q}_v^\times$, $a \neq 1$. Then,*

$$\begin{aligned} J_G^G(\delta, f_v) &= |z|_v^{-1} \int_{\mathbf{K}_v} \int_{\mathbb{Q}_v} f_v \left(k^{-1} \begin{bmatrix} za & y \\ 0 & za^{-1} \end{bmatrix} k \right) dy, \\ J_M^G(\delta, f_v) &= |z|_v^{-1} \int_{\mathbf{K}_v} \int_{\mathbb{Q}_v} f_v \left(k^{-1} \begin{bmatrix} za & y \\ 0 & za^{-1} \end{bmatrix} k \right) \xi_v(z^{-1}(a - a^{-1})^{-1}y) dy \end{aligned}$$

with

$$\xi_v(x) := \begin{cases} \log(1+x^2)^{1/2}, & (v = \infty), \\ \log \max(1, |x|_p^{1/2}) & (v = p < \infty). \end{cases}$$

and

$$J_M^G(z1_2, f_v) = |z|_v^{-1} \int_{\mathbf{K}_v} \int_{\mathbb{Q}_v} f_v(k^{-1} \begin{bmatrix} z & y \\ 0 & z \end{bmatrix} k) \log |y|_v dy$$

Lemma 6.10 (Unipotent orbital integral).

$$I_G^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_v) = \int_{\mathbf{K}_v} \int_{\mathbb{Q}_v} f_v(k^{-1} z \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix} k) dx.$$

6.7. The fine geometric expansion. Let $f = \otimes_v f_v$ and S be as in (6.5).

Proposition 6.11. *Let $\mathfrak{o}$ be the $G(\mathbb{Q})$ -conjugacy class of a $\mathbb{Q}$ -hyperbolic element $\gamma = \text{diag}(a_1, a_2)$ ($a_1, a_2 \in \mathbb{Q}^\times, a_1 \neq a_2$). Suppose S satisfies the condition $a_1, a_2 \in \mathbb{Z}_p^\times$ for $p \notin S$. Then,*

$$J_{\mathfrak{o}}(f) = \frac{1}{2} \sum_{v \in S} \sum_{\delta \in M(\mathbb{Q}) \cap \mathfrak{o}} J_M^G(\delta, f_v) \prod_{w \in S - \{v\}} I_G^G(\delta, f_w).$$

Proposition 6.12. *Let $fo(z)$ ($z \in \mathbb{Q}^\times$) be the unipotent class. Suppose S satisfies the condition $z \in \mathbb{Z}_p^\times$ for all $p \notin S$. Then,*

$$\begin{aligned} J_{\mathfrak{o}}(f) &= \text{vol}([G]) \prod_{v \in S} I_G^G(z1_2, f_v) \\ &+ \frac{1}{2} \sum_{v \in S} J_M^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_v) \cdot \prod_{w \in S - \{v\}} I_G^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_w) + \prod_{v \in S} I_G^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_v). \end{aligned}$$

Proposition 6.13. *Let $\mathfrak{o}$ be a $\mathbb{Q}$ -elliptic conjugacy class. Suppose S satisfies the condition $\text{tr}(\gamma) \in \mathbb{Z}_p$ and $\det(\gamma) \in \mathbb{Z}_p^\times$.*

6.8. Invariant trace formula with the fine geometric expansion. (cf. the introduction of [2]). The weighted orbital integrals are indeed measures on $G(\mathbb{Q}_o)$ but are not $G(\mathbb{Q}_v)$ -invariant. This means that the distribution $J_{\mathfrak{o}}(f)$ is not $G(\mathbb{A})$ -invariant if $\mathfrak{o}$ is $\mathbb{Q}$ -hyperbolic or unipotent. Similarly, on the spectral side, the distribution $J_\chi(f)$ is not $G(\mathbb{A})$ -invariant if $\chi = (P, [\mu]) \in \mathfrak{X} - \mathfrak{X}^G$. Given the fact that $\text{trace} R_0(f)$ is $G(\mathbb{A})$ -invariant, the equality (6.3) suggests that the $G(\mathbb{A})$ -invariance can emerge when individually non-invariant terms $J_\chi(f)$ and $J_{\mathfrak{o}}(f)$ are combined in some way. Actually, this happens very nicely.

6.8.1. The weighted character. Let $\psi = \otimes_v \psi_v : \mathbb{A}/\mathbb{Q} \rightarrow \mathbb{C}^1$ be the unique character such that $\psi(x) = e^{2\pi i x}$ for $x \in \mathbb{Q}_\infty = \mathbb{R}$. For an idele class character $\eta = \otimes_v \eta_v : \mathbb{A}^\times / \mathbb{R}_{>0} \mathbb{Q}^\times \rightarrow \mathbb{C}^1$, let $L(s, \eta) = \prod_v L(s, \eta_v)$ ($\text{Re}(s) > 1$) denote the (completed) Hecke L -function. Recall that it has a meromorphic continuation to $\mathbb{C}$ admitting a possible simple at $s = 0, 1$ satisfying the functional equation

$$L(s, \eta) = \varepsilon(s, \eta) L(1 - s, \eta^{-1})$$

with $\varepsilon(s, \eta)$ the global ε -factor, which is a product of local ε -factors $\varepsilon(s, \eta_v, \psi_v)$.

Definition 15. Let $\mu = (\mu_1, \mu_2)$ be a pair of idele class characters. Define the normalized intertwining operator $R(\mu, s) : \mathbf{H}_\mu^0 \rightarrow \mathbf{H}_\mu^0$ by

$$R(\mu, s)\varphi := m(\mu, s)^{-1}M(s)\varphi, \quad m(\mu, s) := \varepsilon(s, \mu_1\mu_2^{-1}) \frac{L(s+1, \mu_1\mu_2^{-1})}{L(s, \mu_1\mu_2^{-1})}$$

for $\varphi \in \mathbf{H}_\mu^0$.

We have

$$(6.7) \quad \text{tr}(R(\mu, it)^{-1}R'(\mu, it)\pi_{it}(f)|_{\mathbf{H}_\mu^0}) = \text{tr}(M(it)^{-1}M'(it)\pi_{it}(f)|_{\mathbf{H}_\mu^0}) - \frac{m'(\mu, it)}{m(\mu, it)} \text{tr}(\pi_{it}(f)|_{\mathbf{H}_{\mu_0}}).$$

The following is the key lemma.

Lemma 6.14.

$$\check{\phi}_M(f) : m \mapsto \frac{1}{4\pi} \int_{\widehat{M(\mathbb{A})}^1} \int_{\mathbb{R}} \text{tr}(R(\mu, it)^{-1}R'(\mu, it)\pi_{it}(f)|_{\mathbf{H}_\mu^0}) \bar{\mu}(m) d\mu dt$$

on $M(\mathbb{A})^1$ is smooth and is of compactly supported. We have

$$(6.8) \quad \sum_{\mu} \frac{1}{4\pi} \int_{\mathbb{R}} \text{tr}(R(\mu, it)^{-1}R'(\mu, it)\pi_{it}(f)|_{\mathbf{H}_\mu^0}) dt = \sum_{\delta \in M(\mathbb{Q})} \check{\phi}_M(f; \delta).$$

Proof. We include a proof. For a place v , let $\mu_v = (\mu_{1,v}, \mu_{2,v})$ and $\mathbf{H}_{\mu_v}$ the representation space of principal series $(\mu_{1,v} \boxtimes \mu_{2,v})^{G(\mathbb{Q}_v)}$ (see Definition ??). For $(t, s) \in \mathbb{C}^2$, define a $G(\mathbb{A})$ -action $\pi_{(t,s)}$ on $\mathbf{H}_\mu$ by $\pi_{(t,s)}(g) = \pi_{it}(g)|\det g|_{\mathbb{A}}^s$. Then, $(\pi_{(t,s)}, \mathbf{H}_\mu)$ is the same as $(\mu_1| \cdot |_{\mathbb{A}}^{(t+s)/2} \boxtimes \mu_2| \cdot |_{\mathbb{A}}^{(-t+s)/2})^{G(\mathbb{A})}$. We have

$$\begin{aligned} \check{\phi}_M(f; m) &= \frac{1}{4\pi} \int_{\widehat{M(\mathbb{A})}^1} \int_{\mathbb{R}} \int_{\mathbb{R}} \text{tr}(R(\mu, it)^{-1}R'(\mu, it)\pi_{(it, is)}(f)|_{\mathbf{H}_\mu^0}) dt ds d\mu \\ &= \frac{1}{4\pi} \int_{\widehat{M(\mathbb{A})}} \text{tr}(R(\nu)^{-1}R'(\nu)\pi_\nu(f)) d\nu, \end{aligned}$$

where $R(\nu) := R(\mu, it)$ if $\nu = (\mu_1| \cdot |_{\mathbb{A}}^{i(t+s)/2}, \mu_2| \cdot |_{\mathbb{A}}^{i(-t+s)/2})$. Similarly, we define a $G(\mathbb{Q}_v)$ -action $\pi_{(t,s),v}$ on $\mathbf{H}_{\mu_v}$ so that the natural isomorphism $\mathbf{H}_\mu^0 \cong \bigotimes_v \mathbf{H}_{\mu_v}^0$ is a $G(\mathbb{A})$ -isomorphism. Then we define the normalized local intertwining operator $R(\mu_v, s) : \mathbf{H}_{\mu_v}^0 \rightarrow \mathbf{H}_{\mu_v}^0$ by $R(\mu_v, s) := m(\mu_v, s)^{-1}M(\mu_v, s)$, where $m(\mu_v, s)$ is the Euler v -factor of $m(\mu, s)$ and $M(\mu_v, s)$ is the local intertwining operator defined as the analytic continuation of

$$M(\mu_v, s)\varphi_v(g) := \delta_P(g)^{-s/2} \int_{N(\mathbb{Q}_v)} \varphi_v^{(s)}(w_0 n g) dn, \quad \varphi_v \in \mathbf{H}_{\mu_v}^0 \quad (\text{Re}(s) > 0).$$

If φ corresponds to a pure tensor $\bigotimes_v \varphi_v$ in $\bigotimes_v \mathbf{H}_{\mu_v}^0$ with $\varphi_v \in \mathbf{H}_{\mu_v}^0$, then

$$(6.9) \quad R(\mu, s)\phi(g) = \prod_v R(\mu_v, s)\phi_v(g_v)$$

If p is a prime such that $\mu_{1,p}$ and $\mu_{2,p}$ are both unramified, there exists a unique $\mathbf{K}_p$ -invariant vector $\varphi_{\mu_p}^\circ$ in $\mathbf{H}_{\mu_p}^0$ such that $\varphi_{\mu_p}^\circ(1_2) = 1$; then, by a well-known computation of local intertwining operator on $\varphi_{\mu_p}^\circ$, we have $M(\mu_p, s)\varphi_{\mu_p}^\circ = m(\eta_p, s)\varphi_{\mu_p}^\circ$, or equivalently $R(\mu_v, s)\varphi_{\mu_p}^\circ = \varphi_{\mu_p}^\circ$. This happens at all p outside a large finite set S of primes in the factor of the product in (6.9). Other than this, the important properties of the local normalized intertwining operator are summarized as follows:

- (i) For any $\phi_v \in \mathbf{H}_{\mu_v}^0$, the function $s \mapsto R(\mu_v, s)\phi_v$ is entire on $\mathbb{C}$.
- (ii) We have that $R(\mu_v, s) : \mathbf{H}_{\mu_v}^0 \rightarrow \mathbf{H}_{\mu_v}^0$ preserves the inner product if $s \in i\mathbb{R}$.

In particular, the logarithmic derivative $R(\mu_v, -s)R'(\mu_v, s)$ is *regular on* $\text{Re}(s) = 0$. By taking the logarithmic derivative, we get a finite expression from (6.9):

$$R(\mu, it)^{-1}R'(\mu, it)\varphi(g) = \sum_{v \in S} R(\mu_v, -it)R'(\mu_v, it)\varphi_v(g_v) \times \prod_{w \in S - \{v\}} \varphi_w(g_w), \quad t \in \mathbb{R}$$

From this,

$$\text{tr}(R(\nu)^{-1}R'(\nu)\pi_\nu(f)) = \sum_{v \in S} \text{tr}(R(\nu_v)^{-1}R'(\nu_v)\nu_v^{G(\mathbb{Q}_v)}(f_v)) \times \prod_{w \in S - \{v\}} \widehat{f}_w(\nu_v^{G(\mathbb{Q}_w)})$$

for $\nu = \otimes_v \nu_v \in \widehat{M(\mathbb{A})}$. Thus,

$$(6.10) \quad \check{\phi}_M(f; m) = \frac{1}{4\pi} \sum_{v \in S} \int_{\widehat{M(\mathbb{Q}_v)}} \text{tr}(R(\nu_v)^{-1}R'(\nu_v)(\nu_v^{G(\mathbb{Q}_v)}(f_v))\bar{\nu}_v(m_v) d\nu_v \\ \times \prod_{w \in S - \{v\}} \int_{\widehat{M(\mathbb{Q}_w)}} \widehat{f}_w(\nu_w^{G(\mathbb{Q}_w)})\bar{\nu}_w(m_w) d\nu_w.$$

By the descent formula (cf. Lemma 4.1), we have

$$\widehat{f}_w(\nu_w^{G(\mathbb{Q}_w)}) = \widehat{f_{w,P}}(\nu_w) = \int_{M(\mathbb{Q}_w)} f_{w,P}(m_w)\nu_w(m_w) dm_w.$$

Hence, by the Fourier inversion on $M(\mathbb{Q}_w)$,

$$\int_{\widehat{M(\mathbb{Q}_w)}} \widehat{f}_w(\nu_w^{G(\mathbb{Q}_w)})\bar{\nu}_w(m_w) d\nu_w = f_{w,P}(m_w).$$

Since $f_w \in C_c^\infty(G(\mathbb{Q}_w))$, this function has a compact support C_w on $M(\mathbb{Q}_w)$. Since $M(\mathbb{A})^1 \cap (M(\mathbb{Q}_v) \times \prod_{w \in S - \{v\}} C_w)$ is compact subset of $M(\mathbb{A})^1$, we see that the support of $\check{\phi}_M(f)$ is a compact subset of $M(\mathbb{A})^1$.

Formula (6.8) is obtained by the Poisson summation formula applied to the function $\check{\phi}_M(f)$ on the locally compact abelian group $M(\mathbb{A})^1 = \mathbb{A}^1 \times \mathbb{A}^1$ and to the lattice $M(\mathbb{Q}) = \mathbb{Q}^\times \times \mathbb{Q}^\times$ of $M(\mathbb{A})^1$. $\square$

The invariant trace formula (the main idea):

Subtract (6.8) from (6.4) to form

$$\sum_{\chi \in \mathfrak{X}^G} J_\chi(f) + \sum_{\chi \in \mathfrak{X} - \mathfrak{X}^G} \{J_\chi(f) - \phi_M(\chi; f)\} = \sum_{\mathfrak{o}: \mathbb{Q}\text{-ell}} J_{\mathfrak{o}}(f) + \sum_{\mathfrak{o}: \text{non } \mathbb{Q}\text{-ell.}} \{J_{\mathfrak{o}}(f) - \check{\phi}_M(\mathfrak{o}; f)\}$$

where $\chi = (P, [\mu])$ with $\mu = (\mu_1, \mu_2)$, we set

$$\phi_M(\chi; f) := \frac{1}{4\pi} \int_{\mathbb{R}} \text{tr}(R(\mu, it)^{-1}R'(\mu, it)\pi_{it}(f)|_{\mathbf{H}^0(\chi)}) dt$$

with $\mathbf{H}^0(\chi) := \mathbf{H}_\mu^0 \oplus \mathbf{H}_\mu^0$ if $\mu_1 \neq \mu_2$ and $\mathbf{H}^0(\chi) := \mathbf{H}_\mu^0$ if $\mu_1 = \mu_2$, and for hyperbolic $\mathfrak{o} = \{\delta\}_{G(\mathbb{Q})}$ with $\delta \in M(\mathbb{Q}) - Z(\mathbb{Q})$, we set $\check{\phi}_M(\mathfrak{o}; f) := \check{\phi}_M(f; \delta)$, for unipotent class $\mathfrak{o}(z)$ ($z \in \mathbb{Q}^\times$), we set $\check{\phi}_M(\mathfrak{o}; f) := \check{\phi}_M(f, z1_2)$.

Definition 16. For $\chi = (P, [\mu])$, set $I_\chi(f) := J_\chi(f) - \phi_M(\chi; f)$, for $\chi = (G, \pi)$, we set $I_\chi(f) = J_\chi(f)$.

For non elliptic class $\mathfrak{o}$, we set $I_{\mathfrak{o}}(f) := J_{\mathfrak{o}}(f) - \check{\phi}_M(\mathfrak{o}; f)$; for an elliptic $\mathfrak{o}$, we set $I_{\mathfrak{o}}(f) = J_{\mathfrak{o}}(f)$.

Lemma 6.15. The functional $I_\chi(f)$ and $I_{\mathfrak{o}}(f)$ are all $G(\mathbb{A})$ -invariant.

Proof. For $\chi = (P, [\mu])$, the claim is evident due to the identity

$$I_\chi(f) = \frac{1}{4\pi} \int_{\mathbb{R}} \frac{m'(\mu, it)}{m(\mu, it)} \text{tr}(\pi_{it}(f)|_{\mathbf{H}^0(\chi)}) dt.$$

which is proved by (6.7). $\square$

6.9. An invariant avatar of the weighted orbital integral.

Definition 17. Let v be a place and $f_v \in C_c^\infty(G(\mathbb{Q}_v))$. For $\delta \in M(\mathbb{Q}_v) - Z(\mathbb{Q}_v)$, define

$$I_M^G(\delta, f_v) := J_M^G(\delta, f_v) - \frac{1}{4\pi} \int_{\widehat{M(\mathbb{Q}_v)}} \text{tr}(\mathbf{R}(\nu_v)^{-1} \mathbf{R}'(\nu_v) \nu_v^{G(\mathbb{Q}_v)}(f_v)) \bar{\nu}_v(\delta) d\nu_v.$$

For $z \in \mathbb{Q}^\times$, define

$$I_M^G(z1_2, f_v) := J_M^G(z1_2, f_v) - \frac{1}{4\pi} \int_{\widehat{M(\mathbb{Q}_v)}} \text{tr}(\mathbf{R}(\nu_v)^{-1} \mathbf{R}'(\nu_v) \nu_v^{G(\mathbb{Q}_v)}(f_v)) \bar{\nu}_v(z1_2) d\nu_v.$$

Proposition 6.16. The distributions $f_v \mapsto I_M^G(\delta, f_v)$ and $f_v \mapsto I_M^G(z1_2, f_v)$ are $G(\mathbb{Q}_v)$ -invariant.

Lemma 6.17. Let $\mathfrak{o}$ be a $\mathbb{Q}$ -hyperbolic class. Then,

$$I_{\mathfrak{o}}(f) = \frac{1}{2} \sum_{v \in S} \sum_{\delta \in M(\mathbb{Q}) \cap \mathfrak{o}} I_M^G(\delta, f_v) \prod_{w \in S - \{v\}} I_G^G(\delta, f_w).$$

For a unipotent class $\mathfrak{o}(z)$ ($z \in \mathbb{Q}^\times$),

$$\begin{aligned} I_{\mathfrak{o}}(f) &= \text{vol}([G]) \prod_{v \in S} I_G^G(z1_2, f_v) \\ &+ \frac{1}{2} \sum_{v \in S} I_M^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_v) \cdot \prod_{w \in S - \{v\}} I_G^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_w) + \prod_{v \in S} I_G^G(z \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f_v). \end{aligned}$$

Proof. This follows from Propositions 6.11 and 6.12 and Definition 17, and (6.10) together with the descent formula (cf. Lemma 4.1)

$$I_G^G(\delta, f_w) = (f_w)_P(\delta), \quad w \in S - \{v\}.$$

$\square$

Over the reals, the Fourier transform of $I_M^G(\delta, f_\infty)$ and $I_M^G(z1_2, f_\infty)$ are determined by [5]

7. TOWARDS AN EXPLICIT TRACE FORMULA OF $\mathrm{SL}_3(\mathbb{Z})$: WORK IN PROGRESS

The case $\mathrm{SL}_3(\mathbb{Z})$ actually is very difficult in many ways. First of all, the coarse trace formula (3.2) in this form that we start with for a non-uniform lattice in G_3 is not available in literature; only Arthur's work may provides us with a substitute in the setting of adeles, but *not for free*. To expect that distributions from the trace formula are expanded by the global characters like in the case of orbital integrals, the terms from the trace formula should be at least G -invariant. Fortunately, this is attained by the invariant trace formula by Arthur ([2], [3], [4]), which depends on a test function $\prod_{p \leq \infty} f_p$ on the adèle group $G_3(\mathbb{A})$ such that f_p on $G_3(\mathbb{Q}_p)$ is the characteristic function of $G_3(\mathbb{Z}_p)$ outside a "sufficiently large" set S of primes. Here are main issues, that we solved in [11].

- (b) To move from the adelic picture to the wanted case $\mathrm{SL}_3(\mathbb{Z})$ in the classical picture, the choice $S = \{\infty\}$ is necessary. But, Arthur's work does not tell us wether it is possible or not.
- (c) A new kind of distributions (called the weighted orbital integrals and their invariant avatars) emerges from Arthur's trace formula; it has to be expanded to a superposition of global characters; on the archimedean place, for some regular cases, this has been settled by Werner Hoffmann ([10]). But, singular cases have been left untouched.

New technical issues:

- Fourier transform of the invariant avatar of the weighted orbital integrals on $\mathrm{GL}_3(\mathbb{R})$ for singular orbits.

Expected applications:

- Error term estimate in Weyl's law
- Rigorous numerical proof of Ash-Pollack's conjecture for $\mathrm{SL}_3(\mathbb{Z})$ (?)
- Error term estimate in Deitmar-Hoffmann's Prime Geodesic Theorem for $\mathrm{SL}_3(\mathbb{Z})$.

7.1. Further readings. Arthur's papers are all available at

<https://www.math.toronto.edu/arthur/>

Fortunately (for non-experts like me), a very nice survey article by Arthur on his whole work is available [1]. Regarding to the harmonic analysis on real reductive groups, Knapp's famous text book ([12]) is a very convenient reference.

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